

Tutorial using BEAST v2.6.7

Introduction to phylodynamic models

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This is a tutorial to introduce you to Bayesian phylodynamic inference using BEAST2.

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1 Background

Whereas phylogenetics is concerned with describing the evolutionary history of genetic sequences (the phylogeny), phylodynamics aims to describe the process that generated the phylogeny. Whereas phylogenetics gives us a qualitative description of the data, phylodynamics gives us a quantitative description.

In this tutorial we will use different models of population dynamics to describe how the phylogeny grows over time. These models are used as priors for the tree in the Bayesian analysis (the so-called tree-prior).

This tutorial is adapted from <https://taming-the-beast.org/tutorials/Skyline-plots/> (by Nicola F. Müller and Louis du Plessis), but changed to use a heterochronous dataset and with some extra analyses added on. The dataset and analyses are adapted from Hill et al. (2022).

2 Programs used in this Exercise

BEAST2 - Bayesian Evolutionary Analysis Sampling Trees 2

BEAST2 (<http://www.beast2.org>) is a free software package for Bayesian evolutionary analysis of molecular sequences using MCMC and strictly oriented toward inference using rooted, time-measured phylogenetic trees. This tutorial is written for BEAST v2.7.8 (Bouckaert et al. 2014; Bouckaert et al. 2019).

BEAUti2 - Bayesian Evolutionary Analysis Utility

BEAUti2 is a graphical user interface tool for generating BEAST2 XML configuration files.

Both BEAST2 and BEAUti2 are Java programs, which means that the exact same code runs on all platforms. For us it simply means that the interface will be the same on all platforms. The screenshots used in this tutorial are taken on a Mac OS X computer; however, both programs will have the same layout and functionality on both Windows and Linux. BEAUti2 is provided as a part of the BEAST2 package so you do not need to install it separately.

TreeAnnotator

TreeAnnotator is used to summarise the posterior sample of trees to produce a maximum clade credibility tree. It can also be used to summarise and visualise the posterior estimates of other tree parameters (e.g. node height).

TreeAnnotator is provided as a part of the BEAST2 package so you do not need to install it separately.

Tracer

Tracer (<http://beast.community/tracer>) is used to summarise the posterior estimates of the various parameters sampled by the Markov Chain. This program can be used for visual inspection and to assess convergence. It helps to quickly view median estimates and 95% highest posterior density intervals of the parameters, and calculates the effective sample sizes (ESS) of parameters. It can also be used to investigate potential parameter correlations. We will be using Tracer v1.7.2

FigTree

FigTree (<http://beast.community/figtree>) is a program for viewing trees and producing publication-quality figures. It can interpret the node-annotations created on the summary trees by TreeAnnotator, allowing the user to display node-based statistics (e.g. posterior probabilities). We will be using FigTree v1.4.4.

2.0.1 R / RStudio

We will be using R to analyze the output of the Birth-Death Skyline plot. RStudio provides a user-friendly graphical user interface to R that makes it easier to edit and run scripts. (It is not required to use RStudio for this tutorial).

3 Practical

In this tutorial we will analyse two datasets of SARS-CoV-2 genomes sampled from the United Kingdom. One dataset contains genomes belonging to the Alpha Variant of Concern (VOC), which is also known as the B.1.1.7 Pango lineage. This was likely the first SARS-CoV-2 VOC to emerge, in September 2020 (although it was only the second to be identified, in December 2020, shortly after Beta was identified in South Africa). The other dataset contains background non-Alpha SARS-CoV-2 genomes from the UK, collected over the same time period.

The main aim of this tutorial is to introduce you to using phylodynamic models in BEAST2 to estimate epidemiologically relevant parameters. We will do this by using two widely used models for inferring changes in the population dynamics over time in a well-mixed population, the coalescent Bayesian Skyline plot ([Drummond et al. 2005](#)) and the Birth-death Skyline (BDSKY) ([Stadler et al. 2013](#)).

After completing this tutorial you should be able to:

- Set up coalescent and birth-death skyline models in BEAST2
- Make informed choices to parameterise the models
- Sample from the prior
- Post-process skyline output and visualise the results

3.1 The Data

We use two datasets, Alpha and background. The Alpha dataset was produced by subsampling all Alpha genomes from the UK that were sequenced between 1 August 2020 and 1 February 2021 (the oldest Alpha genome only dates from 20 September 2020). We sampled 7 genomes at random from all genomes sequenced in every epiweek during this period. For epiweeks where fewer than 7 genomes are available we used all available genomes. This resulted in 154 genomes. The background dataset was produced by subsampling all other SARS-CoV-2 genomes from the UK that were sequenced during the same time period, sampling 6 genomes per epiweek. This resulted in 160 genomes.

The genomes were all sequenced by COG-UK and the sequences are freely available on their [website](#) (be careful if you download the full alignment, the dataset contains millions of sequences and takes up a lot of space). The website has been deactivated, but is still available on web archives. These genomes are also all available on NCBI GenBank.

If you cloned the Github repository you should already have the subsampled alignments on your drive. Otherwise, you can download the file from [here](#). Please make sure you download the raw files and that your browser doesn't insert HTML code into the file!

3.1.1 Install BEAST 2 packages

We will use BEAUti to generate the BEAST2 XML files. While the coalescent Bayesian Skyline plot is integrated in the BEAST2 core, we need to install the BDMM-Prime package, which implements the Birth-Death Skyline model. Installation of packages is done using the package manager, which is integrated into BEAUti.

Begin by starting **BEAUti2**.

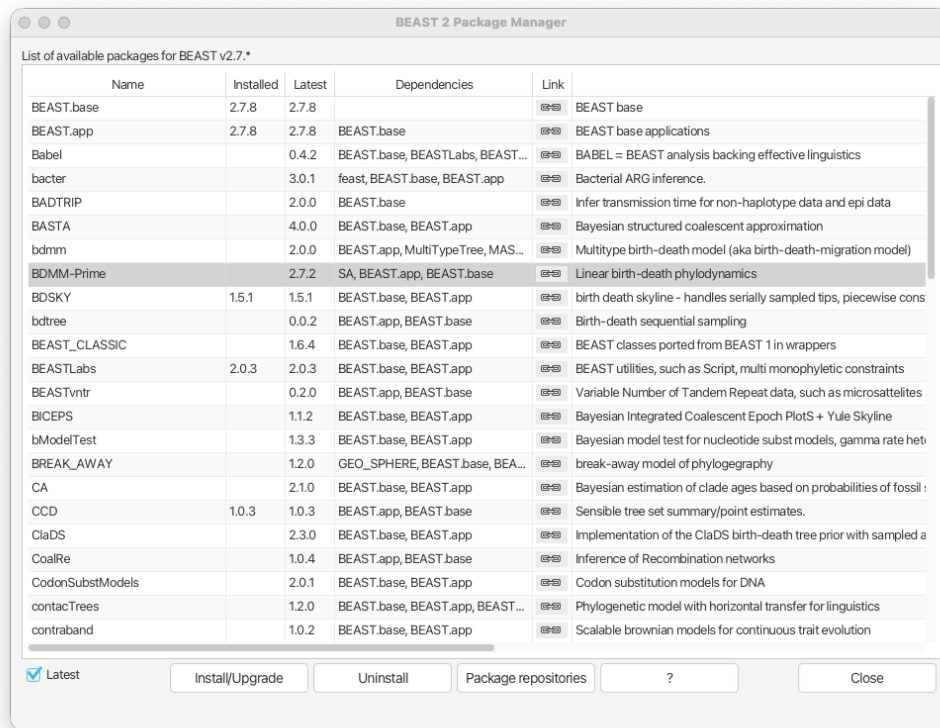


Figure 1: Install the BDMM-Prime package which contains the Birth-Death Skyline model.

Open the **BEAST2 Package Manager** by navigating to **File > Manage Packages**.

Install the **BDMM-Prime** package by selecting it and clicking the **Install/Upgrade** button (Figure 1).

After the installation of a package, the program is on your computer, but BEAUti is unable to load the template files for the newly installed model unless it is restarted. So, let's restart BEAUti to make sure we have the **BDSKY** model at hand.

Close the **BEAST2 Package Manager** and *restart* BEAUti to fully load the **BDMM-Prime** package.

3.2 Setting up a Coalescent Bayesian Skyline analysis

We will set up one analysis file that simultaneously runs the Bayesian skyline model on both the Alpha and the background datasets. In the analysis we will assume that sequences in both datasets evolve under the same site model, but on independent trees with different molecular clock models. We will use a GTR model without rate heterogeneity as the site model and a simple strict clock as the clock model.

Importing the alignments

BEAUti should already be open on the **Partitions** tab.

- Load both alignments (**alpha.fas** and **background.fas**) and set the data type to **nucleotide** for both. Don't split either dataset into any partitions!
- Link the **Site** models, but leave the **Clock** and **Tree** models unlinked.
- Rename the shared **Site Model** to **site**.

Sampling dates

Select the **Tip Dates** tab.

- Select the **alpha** partition and check the **Use tip dates** option.
- Set **Dates specified** to the **as dates with format** option.
- Select yyyy-M-dd from the dropdown box.
- Click the **Auto-configure** button. A window will appear where you can specify how BEAUti can find the collection dates in the sequence headers.
- Select **use everything** and specify **after last |** and click **OK**.

Now repeat the steps above for the **background** partition. (You can also clone the settings as you would for a Site Model, but this doesn't work perfectly and you would still need to make some manual changes).

Site model

Select the **Site Model** tab and select **GTR** in the **Subst Model** drop-down menu. Make sure that the **Substitution Rate** is not being estimated and that the **Gamma Category Count** is set to 0.

Clock model

Select the **Clock Model** tab and check that the clock model is set to **Strict Clock** for both partitions.

Next, we will select the coalescent Bayesian Skyline plot as the tree prior for both trees.

Select the **Priors** tab.

Select **Coalescent Bayesian Skyline** in the drop-down menus next to **Tree.t:alpha** and **Tree.t:background**.

By default the skyline model has a dimension of 5, meaning that the period between the time of the most recent common ancestor (tMRCA) and the most recent sequence is divided into 5 intervals or segments. We would like to set the dimension to 10, to allow some more flexibility.

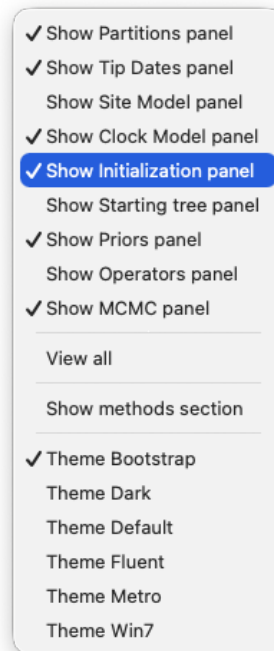


Figure 2: Show the initialization panel.

To change the dimensions of parameters we have to navigate to the **Initialization** panel, which is by default not visible. Navigate to **View > Show Initialization Panel** to make it visible and navigate to it (Figure 2).

Set the dimensions of **bPopSizes.t:alpha**, **bGroupSizes.t:alpha**, **bPopSizes.t:background** and **bGroupSizes.t:background** to 10 each. (the default value is 5) (Figure 3).

Next, we will set a more appropriate prior for the clock rate.

For **clockRate.c:alpha** select **Log Normal** from the drop-down menu

- Expand the options for **clockRate.c:clock** using the arrow button on the left.
- Set the **M** parameter to **5E-4**.
- Set the **S** parameter to **0.2**
- Check the **Mean in Real Space** box

Now repeat the steps above for **clockRate.c:background**.

The BEAUti panel should look as shown in Figure 4.

Finally, we will set the MCMC settings and the output file names.

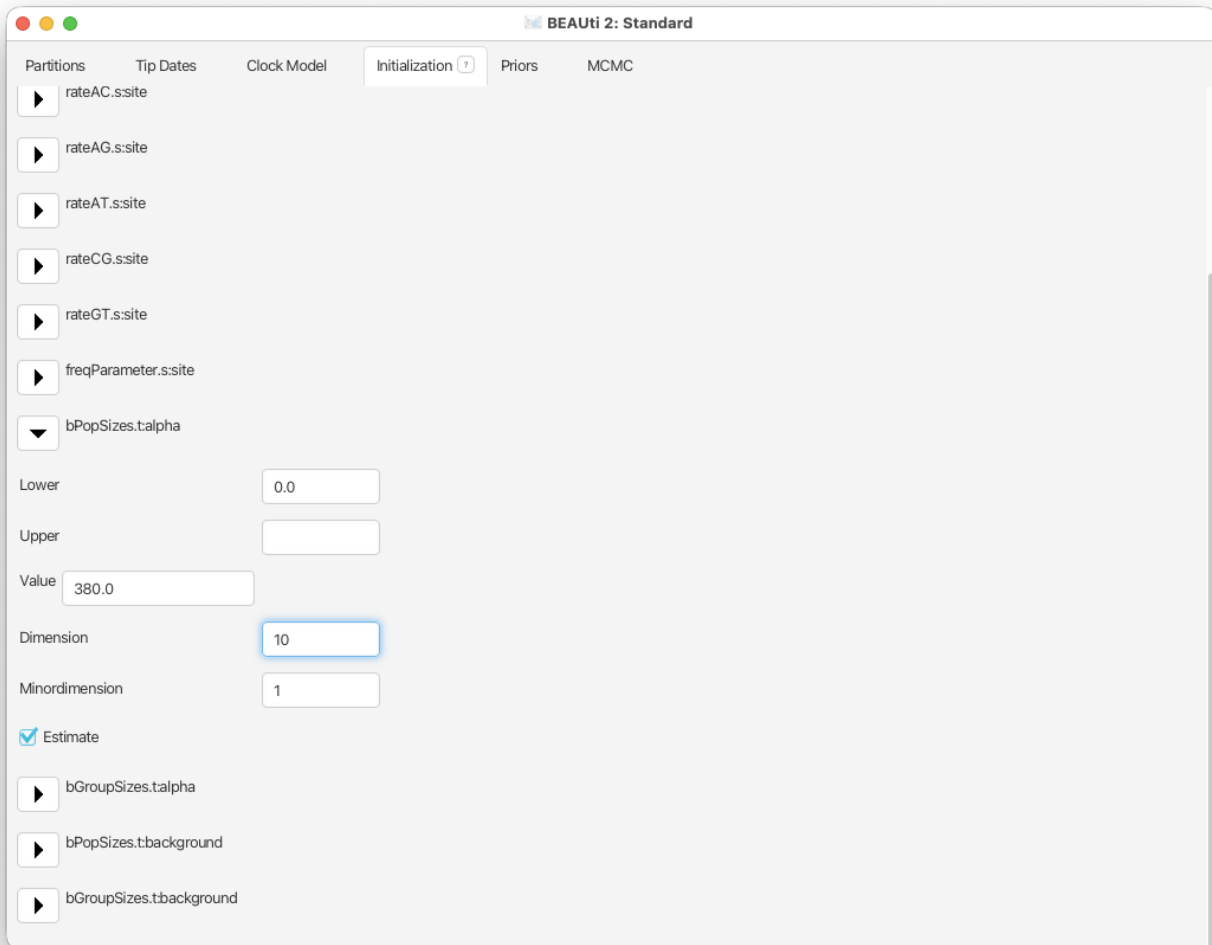


Figure 3: Set the dimension of bPopSizes and bGroupSizes to 10 (remember to do this for both trees).

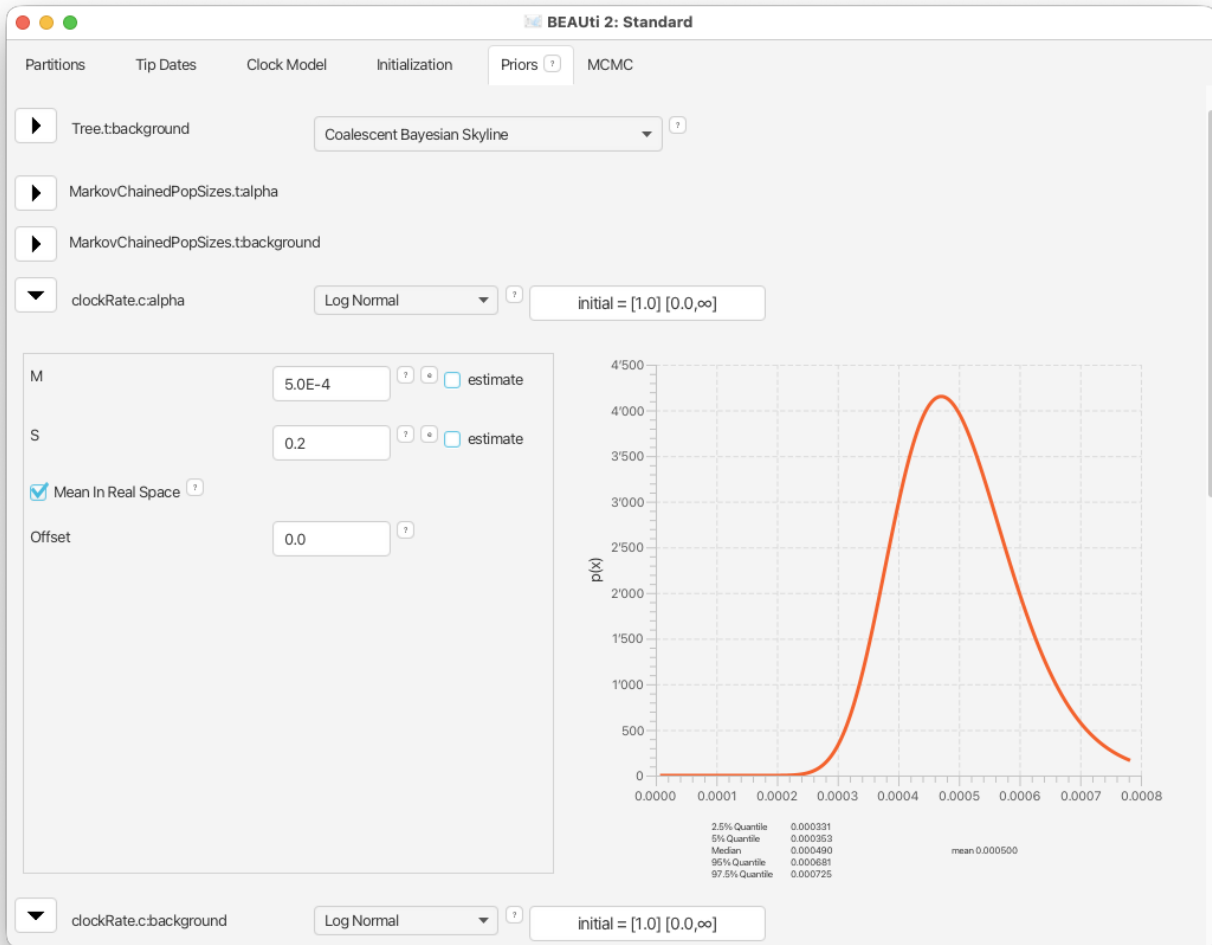


Figure 4: Prior setup.

Go to the **MCMC** tab.

- Set the **Chain Length** to **10'000'000**. (This is the default value)
- Expand the **tracelog** options, change the **File Name** to `$(filebase)_$(seed).log` and set the **Log Every** parameter to **1000**.
- Expand the **screenlog** options and set the **Log Every** parameter to **10'000**.
- Expand the **treelog.t:alpha** options, change the **File Name** to `$(filebase)-$(tree)` and set the **Log Every** parameter to **1000**. (These are the default values)
- Expand the **treelog.t:background** options, change the **File Name** to `$(filebase)-$(tree)` and set the **Log Every** parameter to **1000**. (These are the default values)
- Save the XML file under the name **bsp.xml** using **File > Save**.

When we run the analysis `$(filebase)` in the name of the `*.log` and `*.trees` files will be replaced by the name of the XML file, `$(tree)` by the tree parameter names and `$(seed)` by the random number seed. This is a good idea, since it makes it easy to keep track of which XML files and which random number seeds produced which output files.

When using the coalescent Bayesian Skyline plot it is very important that the **Log Every** parameter for the trace and tree log files are set to the same frequency, or else Tracer won't be able to reconstruct the skyline plot!

Now we are ready to run the analysis. Do **NOT** close BEAUti, as we will return to it in the following sections!

Run the **BEAST2** program.

- Select **bsp.xml** as the **Beast XML File**.
- Set the **Random number seed** to **777** (or pick your favourite number).

Run **BEAST2** by clicking the **Run** button.

The analysis will take some time to complete (allow at least 15-20 minutes). Read through the next section while waiting for your results or start preparing the XML file for the **birth-death skyline** analysis.

3.3 The Coalescent Bayesian Skyline parameterization (optional)

This section is for extra information only and goes into more depth on the model and related it to epidemiological dynamics. It is not necessary for completing the tutorial!

The Kingman coalescent model that the Coalescent Bayesian Skyline is based on assumes that the sequences represent a small sample from a haploid population evolving under Wright-Fisher dynamics (Figure 5). The model works by calculating the probability of the tree under this assumption. This essentially boils down to repeatedly asking the question of how likely it is for two lineages to coalesce (have a common ancestor) in a given time.

The effective population size (N_e) is the inverse of the rate of coalescence λ . The larger N_e is, the less likely lineages are to coalesce. Thus, intervals in a sampled tree with many branching events often coincide

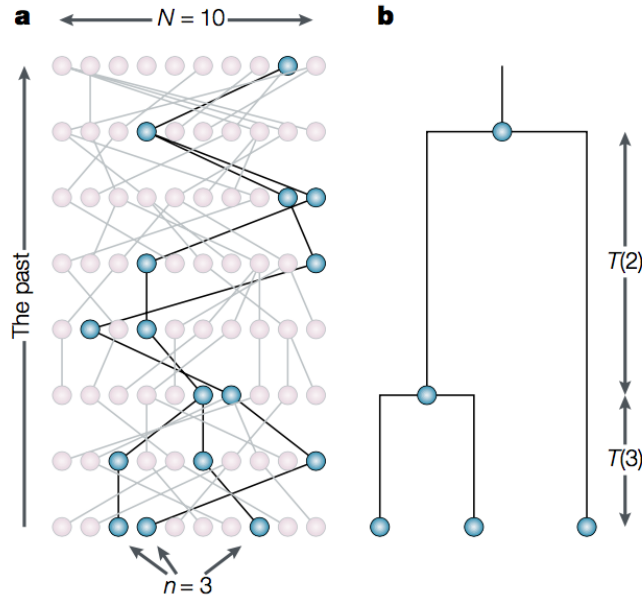


Figure 5: The basic principle behind the coalescent. Figure from (Rosenberg and Nordborg 2002).

with periods when the population size was small. Similarly, few branching events occur during periods of large population size. (Note that these results are conditioned on sampling only a small fraction of the population).

$$\lambda = \frac{1}{N_e} \quad (1)$$

For an SIR model (**S**usceptible, **I**nfectious and **R**ecovered), N_e is proportional to the overall population size N and the number of infected I and inversely proportional to the transmission rate θ .

$$N_e = \frac{I}{\theta} \frac{N}{S} \quad (2)$$

Estimates of N_e therefore do not directly tell us something about the number of infected, nor the transmission rate. However, changes in N_e can be informative about changes in the transmission rate or the number of infected (if they do not cancel out).

The Coalescent Bayesian Skyline model allows N_e to change over time in a nonparametric fashion (i.e. we do not have to specify an equation governing changes in N_e over time). It divides the time between the present and the root of the tree (the tMRCA) into segments, and estimates a different effective population size (N_e) for each segment. The endpoints of segments are tied to the branching times (also called coalescent events) in the tree (Figure 6), and the size of segments is measured in the number of coalescent events included in each segment. The Coalescent Bayesian Skyline groups coalescent events into segments and jointly estimates the N_e (**bPopSizes** parameter in BEAST) and the size (**bGroupSizes** parameter) of each segment. To set the number of segments we have to change the dimension of **bPopSizes** and **bGroupSizes** (note that the dimensions of both parameters always have to be the same). Note that the

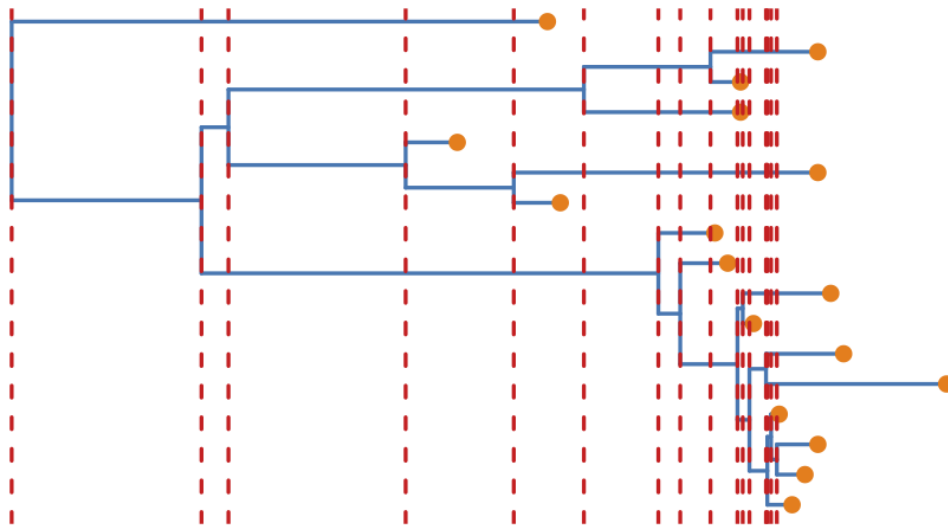


Figure 6: Example tree where the red dotted lines show the time-points of coalescent events.

length of a segment is not fixed, but dependent on the timing of coalescent events in the tree (Figure 6), as well as the number of events contained within a segment (**bGroupSizes**).

Another way to think about the model is as maximally-parameterized, since it infers d change-point times (segment boundaries) and a value for N_e in each segment. This makes the Bayesian Skyline flexible enough to model very complicated N_e dynamics, provided that enough segments are specified. It may be tempting to specify the maximum dimension for the model (each group contains only one coalescent event, thus N_e changes at each branching time in the tree), making it as flexible as possible. This is the parameterization used by the Classic Skyline plot (Pybus et al. 2000), which is the direct ancestor of the Coalescent Bayesian Skyline plot. However, the only informative events used by the Coalescent Bayesian Skyline plot are the coalescent events. Thus, using a maximally-flexible parameterization with only one informative event per segment often leads to erratic and noisy estimates of N_e over time (especially if segments are very short). Grouping segments together leads to smoother and more robust estimates.

Choosing the dimension for the Bayesian Skyline can be rather arbitrary. If the dimension is chosen too low, not all population size changes are captured, but if it is chosen too large, there may be too little information in a segment to support a robust estimate. When trying to decide if the dimension is appropriate it may be useful to consider the average number of informative (coalescent) events per segment. (A tree of n taxa has $n - 1$ coalescences, thus N_e in each segment is estimated from on average $\frac{n-1}{d}$ informative data points). Would this number of random samples drawn from a hypothetical distribution allow you to accurately estimate the distribution? If not, consider decreasing the dimension. There are descendants of the coalescent skyline in BEAST that either estimate the number of segments (Extended Bayesian Skyline (Heled and Drummond 2008)) or do not require the number of segments to be specified (Skyride (Minin et al. 2008) or Skygrid (Gill et al. 2013) models), but instead make strong prior assumptions about smoothly how N_e changes over time.

3.4 Setting up a Birth Death Skyline analysis

We will now repeat the analysis using the Birth Death Skyline model. Either set up a new XML file in BEAUti and follow the same steps as above until you've specified the site and clock models or else simply go back to BEAUti and edit the previous analysis file.

We will need to set the tree prior to **BDMMPri**me. (Note that the Birth Death Skyline model was originally implemented in the **BDSKY** package, which is still available. BDMMPri is a newer, more efficient implementation that makes it easier to specify complex models and can also handle structured populations.)

Select the **Priors** tab.

Select **BDMMPri**me in the drop-down menu next to **Tree.t:alpha**.

Expand the options using the arrow button on the left and select **Epi Parameterization** from the dropdown menu.

Note that we can now specify the settings for the model's parameterization. In the Epi Parameterization we have the **Re** (effective reproduction number), **Become Uninfectious Rate** (the reciprocal of the mean infectious period), **Sampling Proportion** (the fraction of removal events that are sampled and sequenced), **Rho Sampling** (this confusingly named parameter denotes concerted census-sampling events at pre-specified times), **Removal Prob** (the probability that an individual is removed from the infectious population upon sampling), **Migration Rate** (the rate of movement between population groups, if more than one population is specified), and **Re Among Demes** (the effective reproduction number between population groups). For each of these parameters we can also change their dimensions (epochs) and set the change times. There are also options for conditioning the process and setting types. We will ignore most of these here and only focus on the Re, Become Uninfectious Rate and Sampling Proportion.

We will set the dimension of R_e , which denotes the average number of secondary infections caused by an infected person at a given time during the epidemic, i.e. an R_e of 2 would mean that every infected person causes two new infections on average. In other words, an R_e above 1 means that the number of cases are increasing, therefore the disease will cause an exponentially growing epidemic, and an R_e below 1 means that the epidemic will die out. We will also set the dimension of the **samplingProportion** parameters, so that we can also estimate changes in the sampling proportion over time. For both parameters we will assume that change times are spaced equally between the origin of the process and the most recent sample.

In general, we need to fix one of the rates of the birth-death skyline model in order to make the model identifiable, although in some cases we can get away with strong priors on all rates. Here we will fix the **Become Uninfectious Rate** to 36.5, which translates to a mean infected period of 10 days (the inverse of the rate), which we know to be a realistic estimate for SARS-CoV-2.

Set the number of change times for **Re** and **Sampling Proportion** to 9 (so the parameters have dimension 10). For both parameters check **Relative to process length** box and press the **Distribute evenly** button. The change times should now read as 0.1, 0.2, ..., 0.9.

Uncheck **Estimate values** for the **Become Uninfectious Rate** and enter 36.5 as the value under

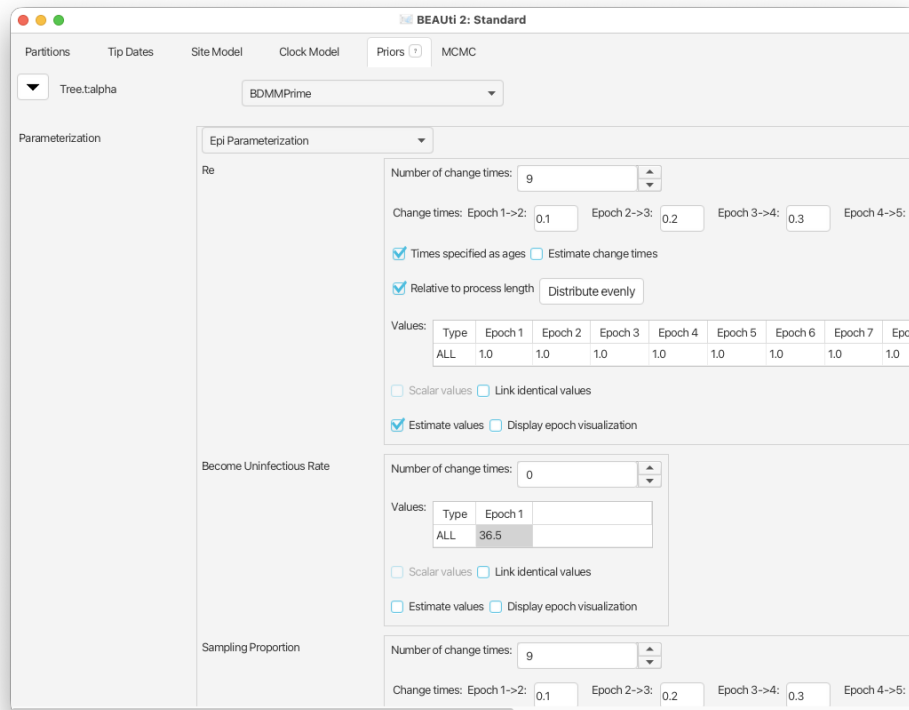


Figure 7: Settings of the BDMM Prime model.

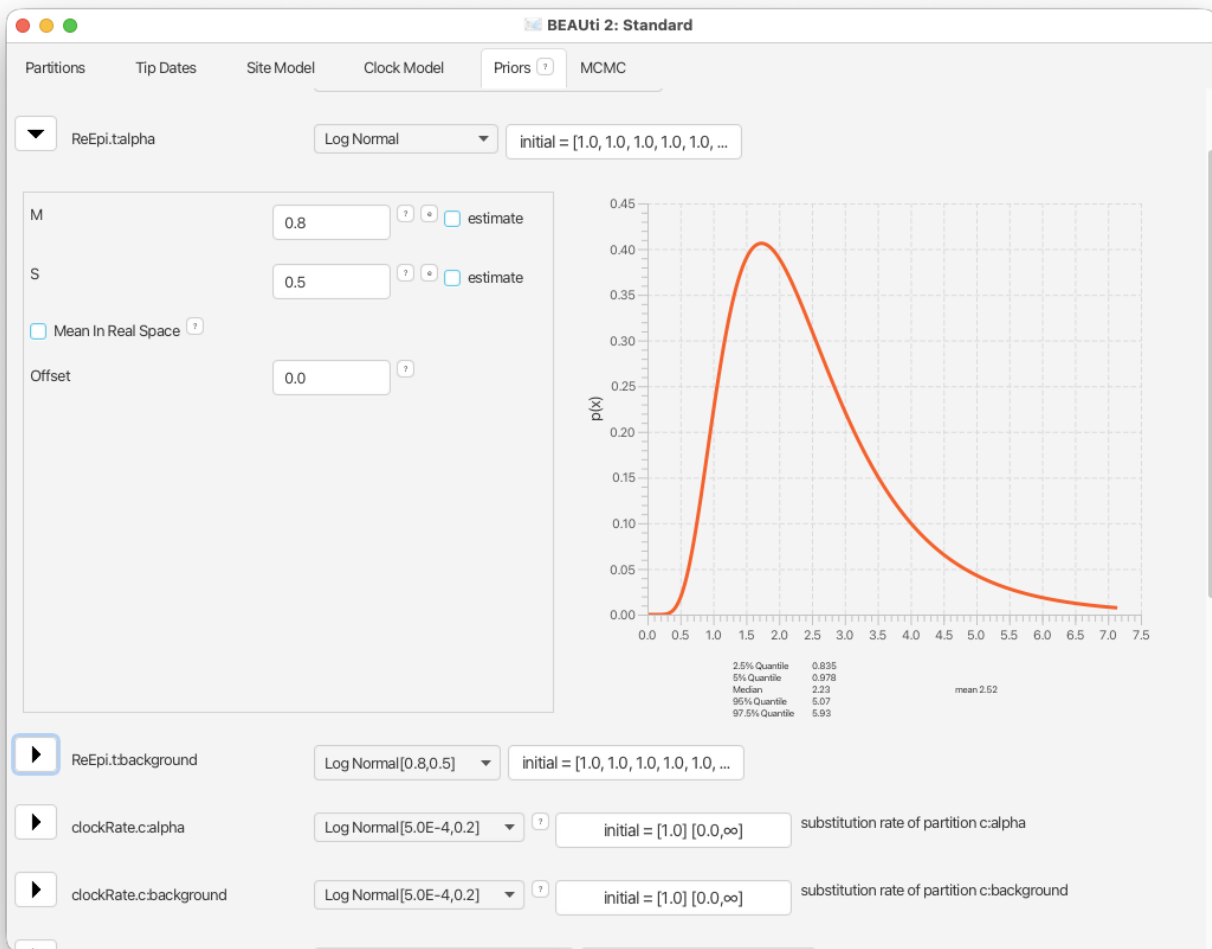
Epoch 1.

Leave the rest of the BDMMPrime parameterisation on the default settings. Your panel should look as in Figure 7.

Now follow the same steps for **Tree.t:background**.

Now R_e and the sampling proportion will be allowed to change at 9 times equally spaced between the origin of the epidemic and the time of the most recent sample, in our case March 1st, 2021. Choosing this dimension can be arbitrary and may require the testing of a few different values. Too few intervals and not all rate shifts are captured. Too many intervals and the intervals may not contain enough information to infer parameters.

We will use a lognormal prior for R_e . This is a good prior distribution to use for rates since it is always positive (a rate cannot be negative) and has a long tail defined over all positive numbers. The long tail allows arbitrarily high estimates of R_e , but does not place much weight on very high rates. This agrees with our prior knowledge about R_e (most diseases have an R_e between 1.2 and 8. Measles is one of the most infectious diseases we know of and has $R_e \approx 18$).

Figure 8: Setting the R_e priors.

Select a **Log Normal** distribution for the **ReEpi.t:alpha** prior.

Click on the arrow to expand the options and set **M** to 0.8 and **S** to 0.5.

Now repeat the above steps for the **ReEpi.t:background** parameter (Figure 8).

This prior has a median above 1 and places most of the weight below 5. This agrees with what we know of the SARS-CoV-2 epidemic in the UK during the second half of 2020 and the start of 2021, when cases grew with the second and third waves of the epidemic. Note that this prior is used for each of the R_e intervals (the Birth-Death Skyline assumes that R_e is independent in each of the intervals).

The sampling proportion represents the proportion of removed lineages during each interval that were sequenced (i.e. included in the alignment). We will use a Beta distribution for the sampling proportion prior. Beta distributions are a very flexible class of distributions that are only defined between 0 and 1,

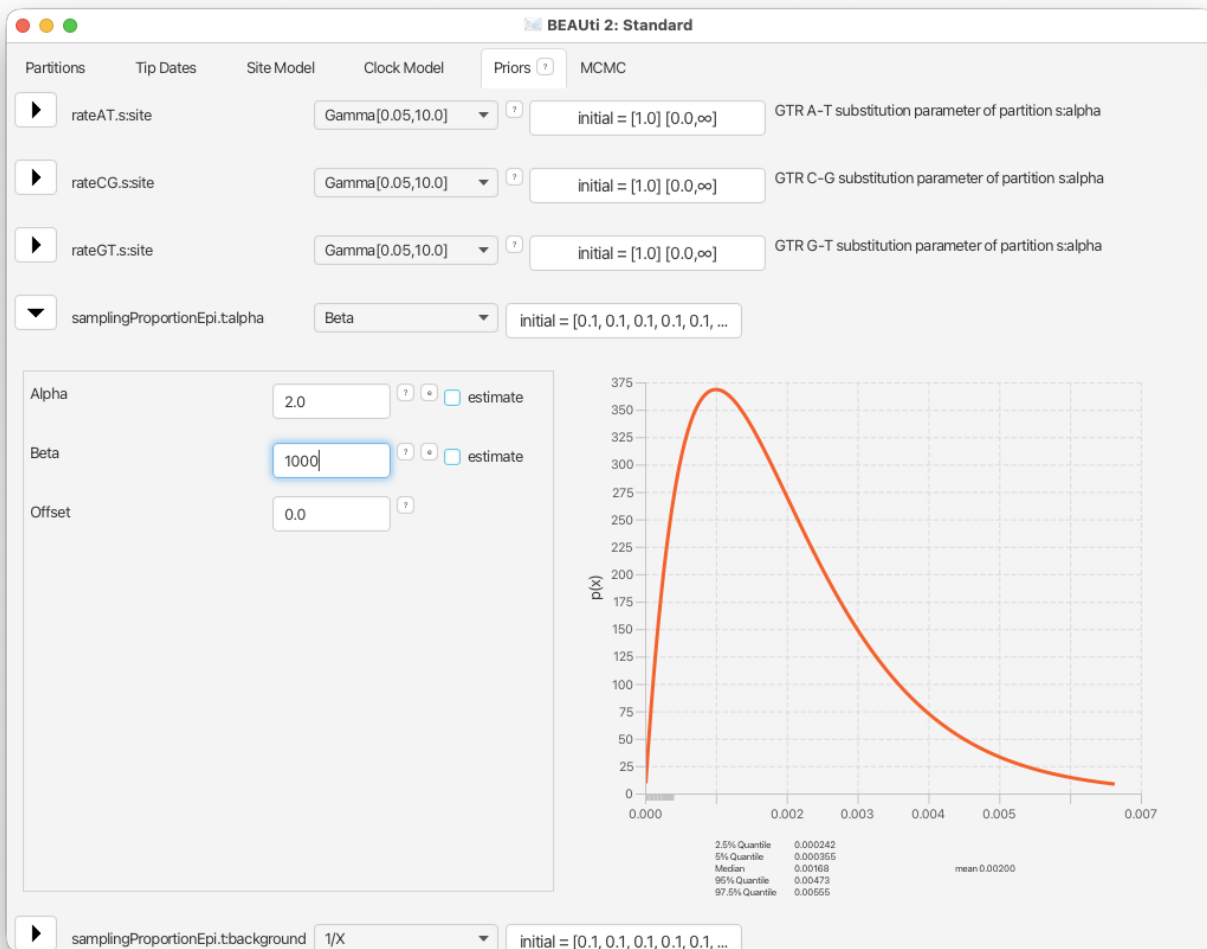


Figure 9: Setting the sampling proportion priors.

making them well-suited to use for proportions. A Beta distribution can be interpreted as modelling the probability of success, out of a number of trials, which is ideal for a sampling model. To decide which parameter values to use for the Beta distribution you can think of $\alpha - 1$ as the number of successes (cases that were sequenced/observed) and $\beta - 1$ as the number of failures (cases that were not sequenced/observed).

Select a **Beta** distribution for the **samplingProportionEpi.t.alpha** prior.

Click on the arrow expand the options and set **Alpha** to **2** and **Beta** to **1000** (Figure 9).

Now repeat the above steps for the **samplingProportionEpi.t.background** prior.

The prior we set assumes we are sequencing roughly 1 out of every 1000 cases (the distribution has a median probability of sequencing a removed lineage of 0.000168).

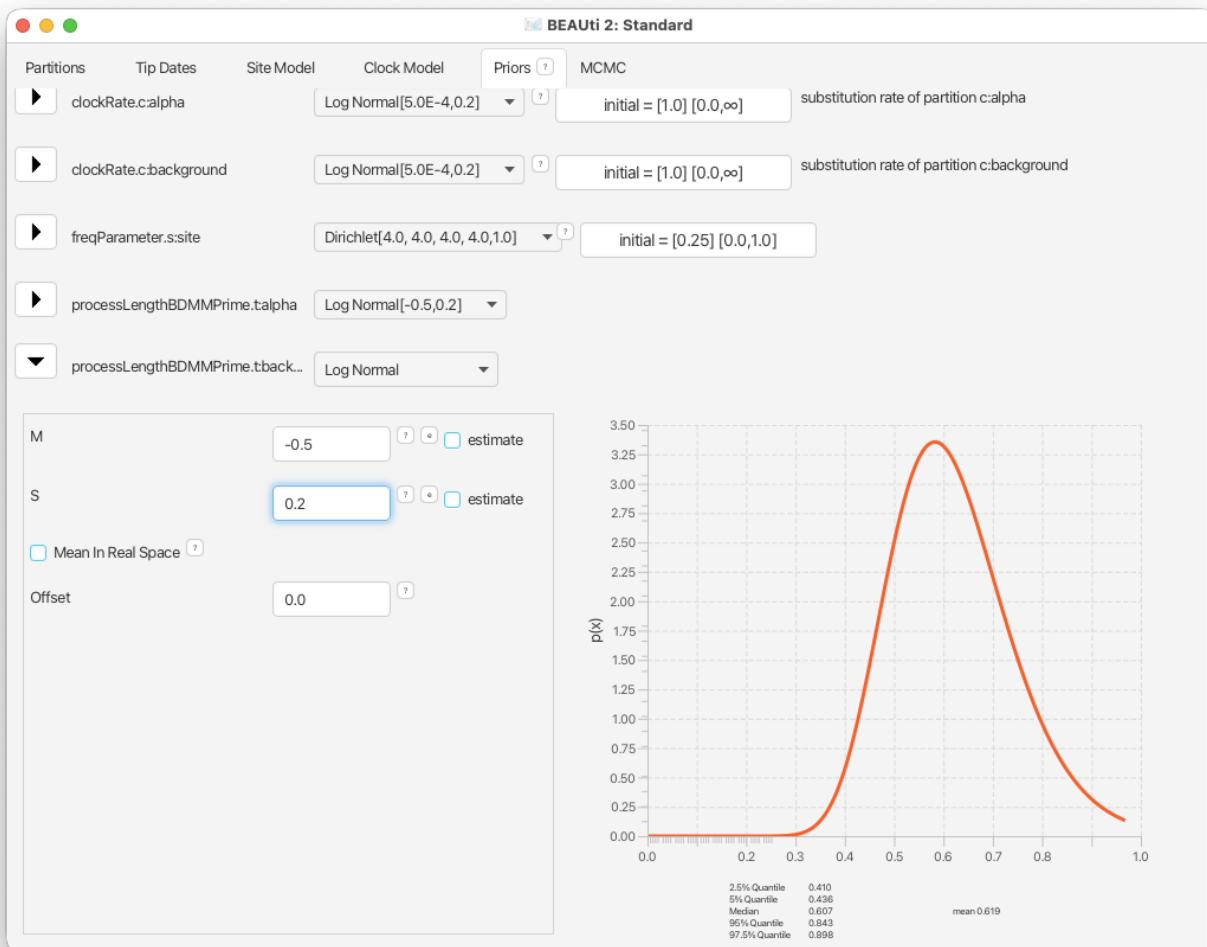


Figure 10: Setting the priors on the origins of the epidemics.

The birth death process has one more parameter we need to look at. The origin defines the time between the start of the process and the most recent sequence in our dataset. Within the **BDMMPriME** package, the origin is called the process length. We will once again use a log normal distribution for this parameter. Note that the origin also has to be positive and needs to be bigger than the MRCA of the tree. At this point in the COVID-19 pandemic it had only been circulating for a little more than a year, thus we want to set a prior that reflects this knowledge. Keep in mind that the origin of the Alpha alignment would be the index case of the Alpha VOC and not of SARS-CoV-2! Similarly, the origin of the background alignment would be the index case that led to all genomes represented in the dataset, which may not be the same as the index case of the entire pandemic.

Set a **Log Normal** prior for both **processLength** parameters with **M = -0.5** and **S = 0.2** (Figure 10).

Double-check that the clock rate priors are the same as those set earlier for the coalescent Bayesian Skyline

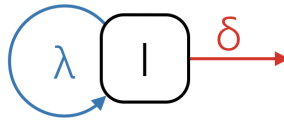


Figure 11: A schematic of a simple birth death model.

plot. The rest of the priors pertain to the site model parameters and we can leave them as they are.

Before we can save the file we need to deactivate a few of the output files that **BDMMP** creates by default. The typed tree and typed node loggers are only useful for structured analyses, but are still activated by default. The trajectory loggers can give an estimate of prevalence over time, but this can be computationally expensive to compute and create massive files. Trajectories can also be inferred *post-hoc*. These loggers should be deactivated by default, but it doesn't hurt to check!

Select the **MCMC** tab.

Leave the options for **tracelog**, **screenlog**, **treelog.t:alpha**, and **treelog.t:background** as for the coalescent Bayesian skyline plot analysis. For each of the other loggers expand the options and uncheck **Enable Logger**.

Now save the file as `bdsky.xml` and run it in BEAST2.

Read through the next section and set up the next XML file while waiting for the analysis to finish.

3.5 The Birth-Death Skyline parameterization (optional)

This section is for extra information only and goes into more depth on the model and related it to epidemiological dynamics. It is not necessary for completing the tutorial!

The birth-death-sampling model is parameterized very differently from the coalescent model, using per lineage rates and an explicit sampling model (whereas the coalescent model conditions on the samples). This makes the birth-death-sampling model more powerful, but also much more complex. A basic birth-death model has a birth rate (λ), the rate at which lineages are added to the tree, and a death rate (δ), the rate at which lineages are removed from the tree (Figure 11). In an infectious disease epidemic λ can be thought of as the transmission rate, the rate at which infected individuals infect susceptibles, while δ can be thought of as the becoming uninfected rate, the rate at which infected individuals recover, die or are isolated. In species tree inferences these rates can be thought of in terms of speciation and extinction.

In addition, the model has an origin parameter. Whereas coalescent models work backward-in-time from the sampled sequences, birth-death models work forward-in-time from the origin. Hence, the model needs an origin time, which can also be jointly estimated along with the other parameters. The origin will always be bigger than the tMRCA of the sampled tree, since the sampled tree is by definition smaller than the complete tree.

The **Birth Death Skyline Serial** model we used was parameterized in terms of R_e and δ . Recall that $R_e > 1$ means that an epidemic will keep growing. We can see this from the definition of R_e as the ratio

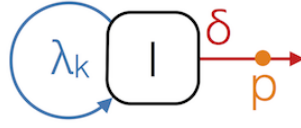


Figure 12: A schematic of the Birth Death Skyline Serial model.

of the birth and death rates.

$$R_e = \frac{\lambda}{\delta} \quad (3)$$

We used this parameterization simply because it is often easier to specify priors for R_e than the transmission rate, and because R_e is often more informative for prevention efforts.

In addition, the model assumes that the data are heterochronous (sampled at different times). It assumes that:

$$\delta = \psi + \mu \quad (4)$$

where ψ is the rate at which lineages are sampled through time and μ is the rate at which lineages are removed from the tree for any other reason (death, recovery, extinction etc.). (In this case the ρ parameter is no-longer available, because samples are collected through time, and not just at one timepoint). By default, the model is parameterized in terms of R_e , δ and p , the sampling proportion:

$$p = \frac{\psi}{\psi + \mu} \quad (5)$$

The sampling proportion is the proportion of all removed lineages that were sampled, and can be used to obtain a rough estimate of the total population size. This model is useful for studying infectious disease dynamics, because samples are often collected over the course of an epidemic.

You can also see that the model **Birth Death Skyline Serial** assumes that upon sampling a lineage is removed from the tree (e.g. in a disease model the sampled individual cannot transmit the disease after sampling). The consequence for the phylogeny is that a sampled lineage cannot be a direct ancestor of any other lineage in the tree. This assumption can be relaxed, but we will not do so during this tutorial.

The Birth-Death Skyline model is very flexible and allows any or all of these rates to change independently over time. This is done by dividing the time from the origin to the most recent sample into dimension d equally spaced intervals (see Figure 13). The rates are then allowed to change between intervals. Since some rates (e.g. λ and δ) are highly correlated, it is not always a good idea to let all rates change over time because it can lead to poor mixing or biased estimates. It is also possible to specify the change-point times more flexibly, or even estimate them.

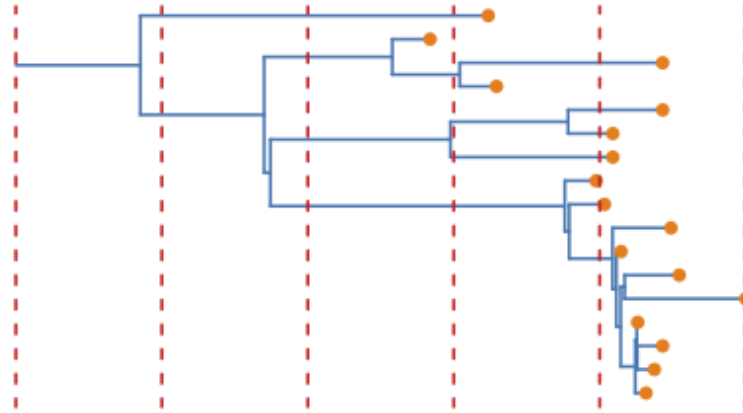


Figure 13: Example tree where the red dotted lines are an example of where rates could be allowed to change on the tree. The branch at the root (compare Figure 6) is indicating the origin of the epidemic, which is also estimated by the birth-death model.

3.6 Analyzing the Coalescent Bayesian Skyline results

Once BEAST2 has finished running the coalescent Bayesian Skyline analysis, open Tracer to get an overview of the BEAST2 output. When the main window has opened, choose **File > Import Trace File...** and select the file called `bsp_777.log` that BEAST2 has created, or simply drag the file from the file manager window into Tracer.

Open **Tracer**. Drag and drop the `bsp_777.log` file into the open Tracer window.

Alternatively, use **File > Import Trace File...** (or press the **+** button below the **Trace Files** panel) then locate and click on `bsp_777.log`.

The Tracer window should look as shown in Figure 14. Because we only ran a short chain for a complicated analysis many parameters have very low ESS values. You can find a log file of an identical analysis that was run for 100 million MCMC steps in the `precooked_runs/` folder (`bsp_long.log`).

Note that many parameters have either `.alpha` or `.background` appended to their names. These are parameters that we inferred independently for the Alpha and background alignments/partitions. The `popSizes` parameters represent the effective population sizes for each alignment in each interval, whereas the `groupSizes` parameters are the sizes of each interval (more specifically, the number of coalescent events within an interval). Since populations naturally grow/decline through exponential growth/decay, it is more natural to visualise the population sizes on a log scale.

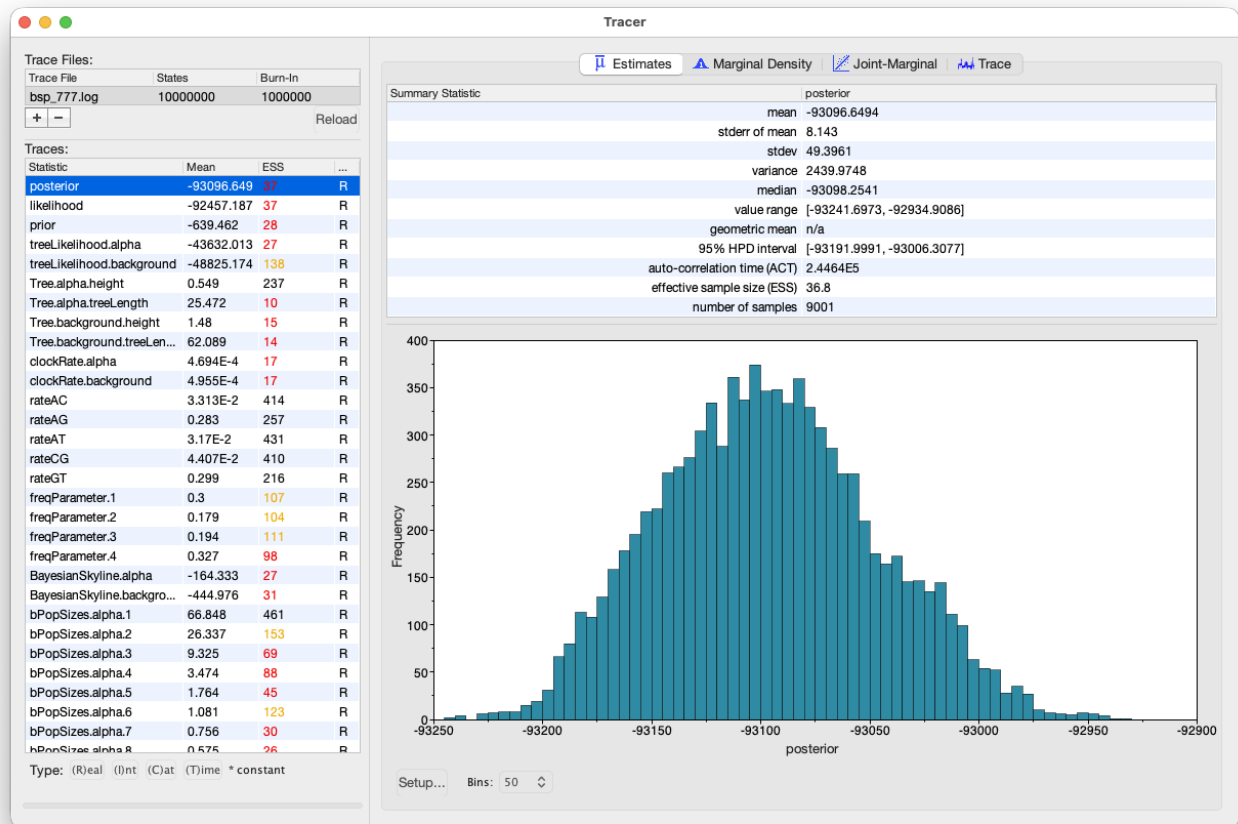


Figure 14: Tracer showing a summary of the BEAST2 run of the coalescent Bayesian Skyline plot with an MCMC chain length of 10'000'000 and no constraints.

Select the **Marginal Density** tab.

- Select all of the **popSizes.alpha** parameters using **shift + click**.
- Open the **Display** drop-down menu and select **Violin**.
- Click on the setup button at the bottom and check **Log axis**.

We see that the effective population size of Alpha appears to have grown linearly on a log scale, which translates to exponential growth. Similarly, the background appears to have first grown exponentially, and then stabilised at a constant population size. On the other hand, the group sizes parameters were not inferred very precisely and there is a large variation in their sizes (Figure 15). This indicates smooth population size changes.

However, these are just indications, and we need to combine both parameters to reconstruct the population dynamics. To do that Tracer needs to know when the coalescent events on each posterior tree was, and therefore also needs the `*.trees` file. This is the reason why it's important to log the trace and tree files at the same frequency for the coalescent Bayesian Skyline plot. But before we can do that, we need to know the decimal date at the time of the most recent sample in our dataset (March 1st, 2021).

Open R or Rstudio.

Type in:

```
library(lubridate)
lubridate::decimal_date(ymd("2021-03-01"))
```

This should give:

```
[1] 2021.162
```

Return to Tracer and navigate to **Analysis > Bayesian Skyline Reconstruction**.

- Choose `bsp-alpha.trees` next to **Trees Log File**.
- Set **Population Size** to `bPopSizes.alpha` and **Group Size** to `bGroupSizes.alpha`.
- Set the **trace of the root height** to `Tree.alpha.height`.
- Check the **Use manual range for bins** checkbox and enter **2020** and **2021.162** for the minimum and maximum times.
- Enter **2021.162** for the **Age of youngest tip**.
- Press **OK** to reconstruct the past population dynamics (Figure 16).

Without closing the resulting plot, go back to the main Tracer window and repeat the same steps for the background alignment.

This should reconstruct the population dynamics for both alignments across the same time period (Figure 17). In general it is not necessary to use a manual range for the bins, but we used it here to ensure

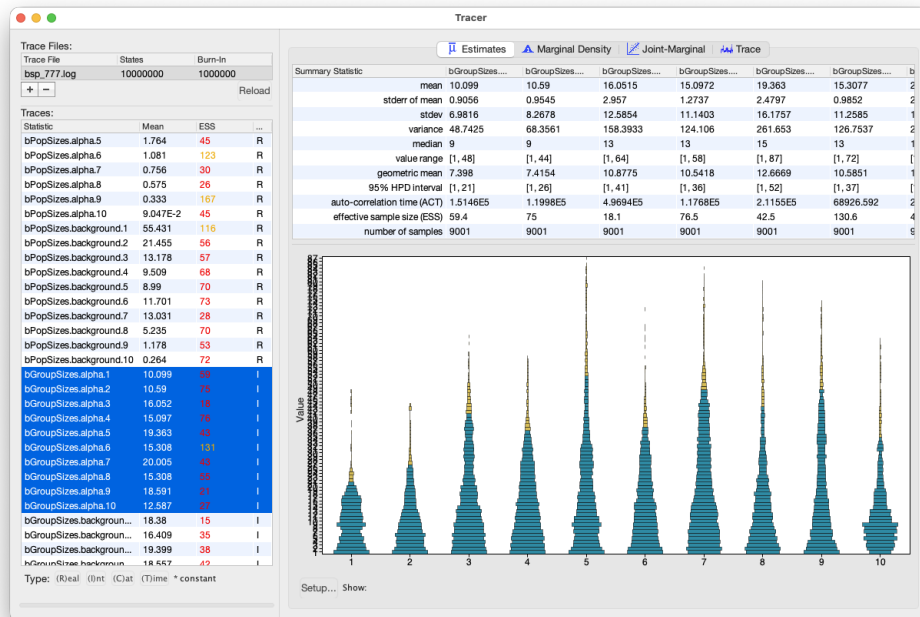
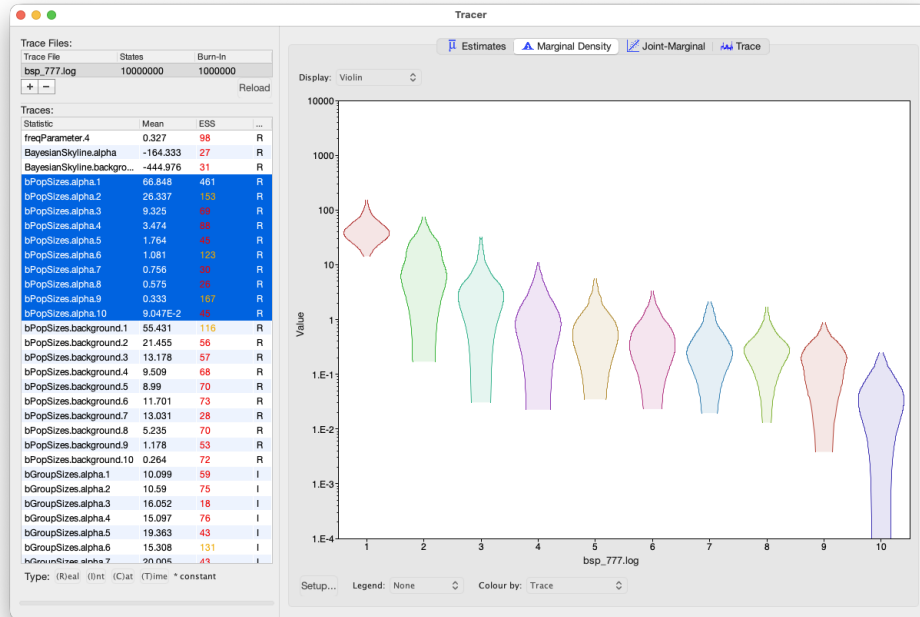


Figure 15: Visualising the population sizes and group sizes in Tracer.

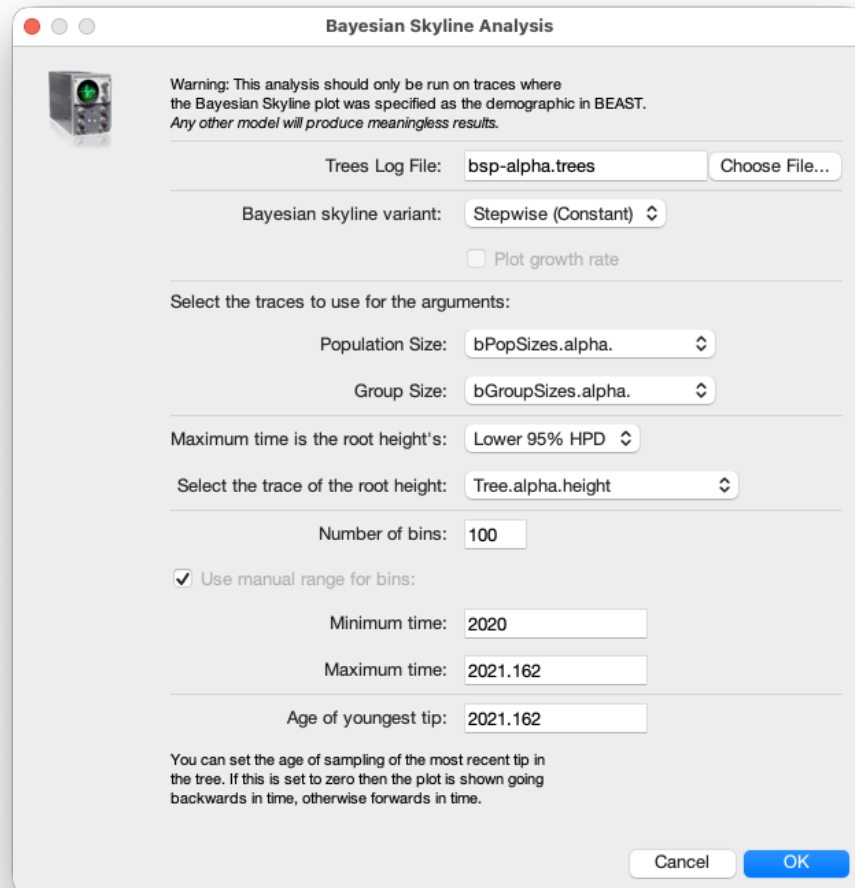


Figure 16: Reconstructing the Bayesian Skyline plot in Tracer.

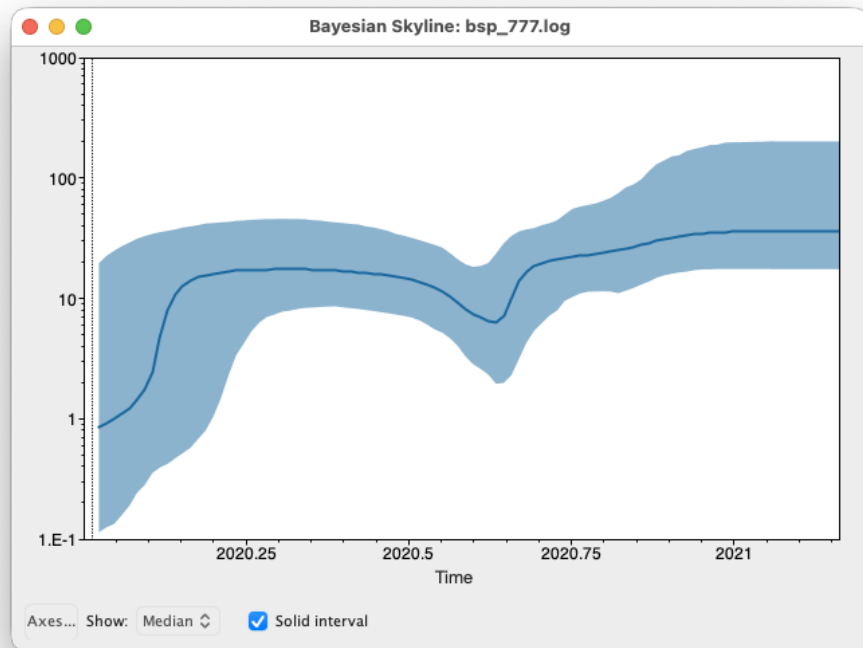
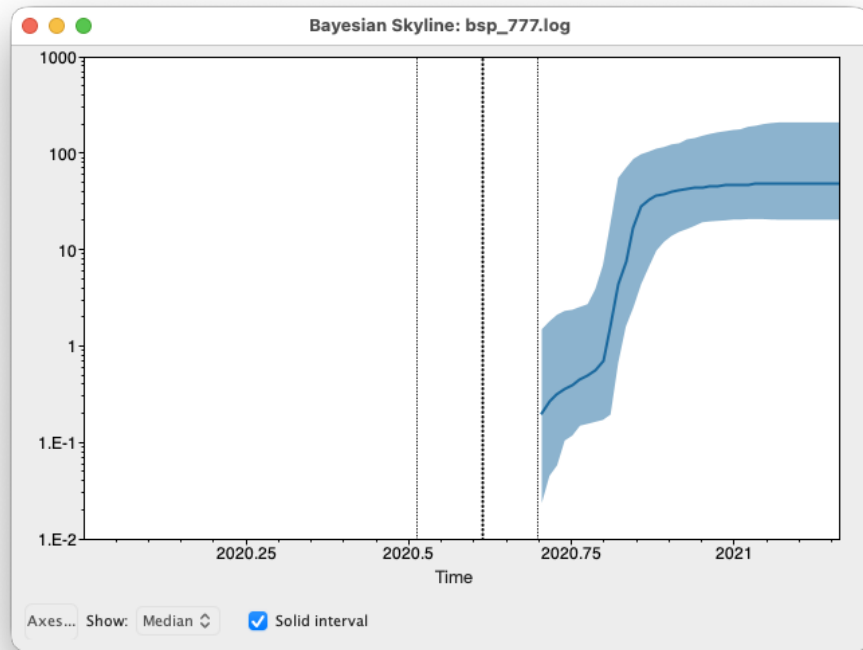


Figure 17: Coalescent Bayesian Skyline analysis output. The thick line is the median estimate of the estimated effective population size (can be changed to the mean estimate). The solid interval represents the upper and lower bounds of the 95% HPD interval. The x-axis is the time in years and the y-axis is on a log-scale.

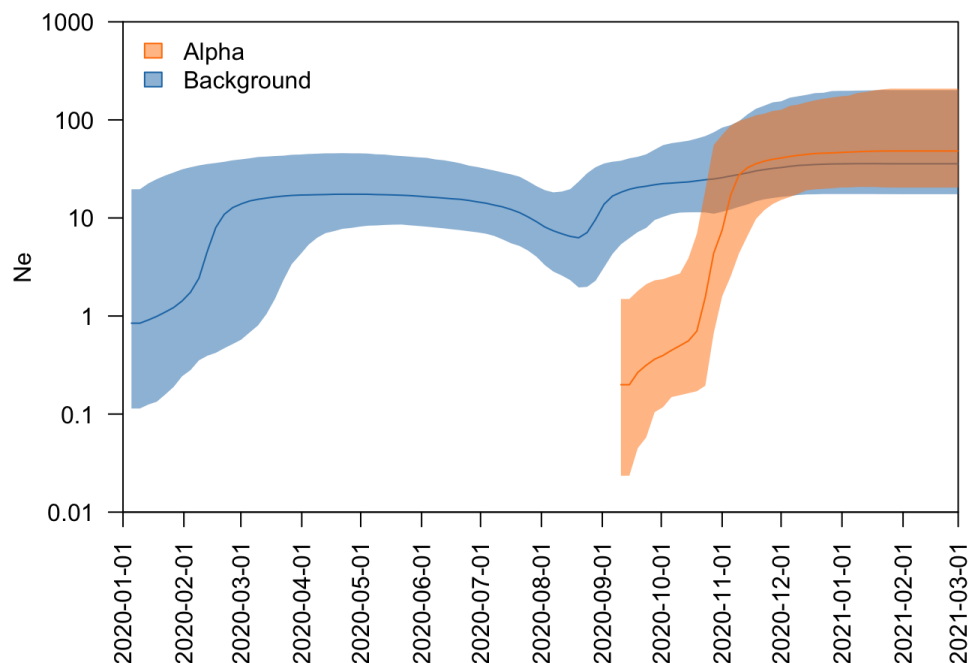


Figure 18: Both Bayesian Skyline Plots on one set of axes.

that both plots are over the same X-axis range and more easily comparable. There are two ways to save the analysis, it can either be saved as a `*.pdf` for display purposes or as a tab delimited file, which can be loaded in R and used for custom plots (Figure 18).

Navigate to **File > Export Data Table**.

Enter the filenames as `bsp_alpha.tsv` `bsp_background.tsv` and save the files.

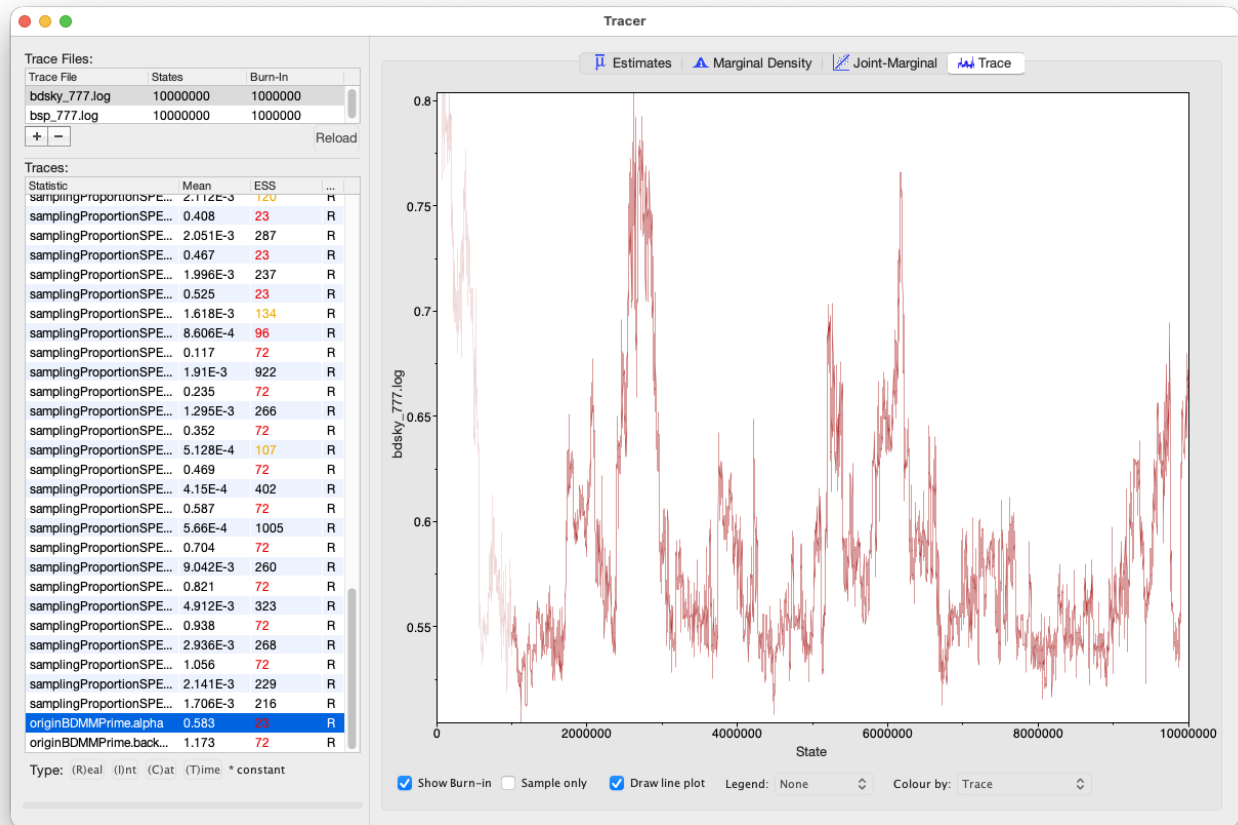


Figure 19: Trace of the origin of the Alpha alignment after 10'000'000 MCMC steps.

3.7 Analyzing the Birth Death Skyline results

If we open the trace file for the Birth Death Skyline analysis we notice that the ESS values are also not good. The trace for the origin of the Alpha alignment is especially bad and highly autocorrelated (Figure 19). Recall that the samples in the trace should approximate uncorrelated draws from the target distribution, which should result in the traces resembling white noise. Obviously we would have to run this analysis much, much, much longer for this parameter to begin mixing well. You can find a log file of an identical analysis that was run for 100 million MCMC steps in the `precooked_runs/` folder (`bdsky_long.log`), but even that analysis still hasn't reached convergence.

There is no equivalent visualization of the skyline plot of a Birth-Death Skyline analysis in Tracer as there is for the Coalescent Bayesian Skyline. But because we parameterised the model to have equally spaced intervals, we can already get an idea of the estimates just by looking at the inferred R_e values (see Figure 20). This gives us a good idea of the trend, but it is not completely accurate. Since we are also estimating the origin, the interval times are slightly different in each posterior sample and overlap slightly. The advantage of this is that we get a smooth estimate through time. The disadvantage is that we need to do some extra post-processing to plot the smooth skyline. We will use R to post-process and plot the Birth Death Skyline. The below steps are also in an RMarkdown notebook in the `scripts/` directory.



Figure 20: Estimated population dynamics for Alpha by the birth-death model in Tracer.

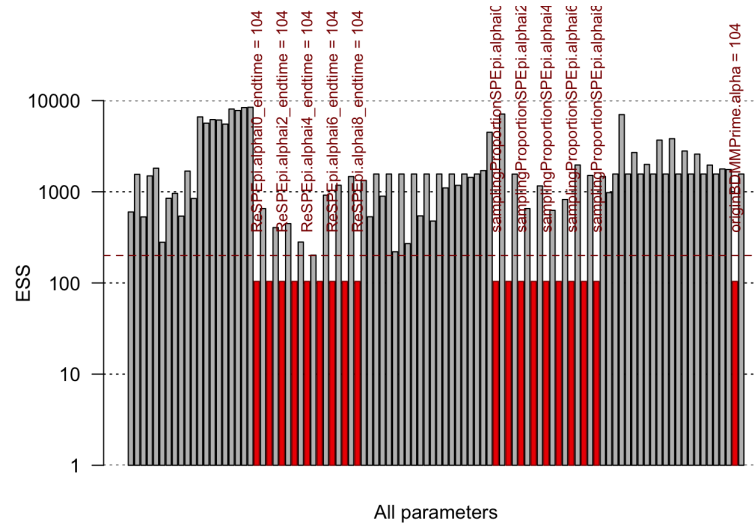


Figure 21: The ESS values of the trace (clearly the analysis needs to be run for longer).

First, install the necessary packages. Once installed, we don't need to install the packages again.

```
install.packages("devtools")
install.packages("lubridate")
install.packages("coda")
install.packages("RColorBrewer")
devtools::install_github("laduplessis/bdskytools")
devtools::install_github("laduplessis/beastio")
```

Now we have to load the packages into our R workspace before we can use the functions in the packages.

```
library(lubridate)
library(coda)
library(bdskytools)
library(beastio)
library(RColorBrewer)
```

To plot the results, we need to first tell R where to find the `*.log` file of our run and then load it into R (discarding 10% of samples as burn-in). If you are using RStudio, you can change the working directory to the directory where you stored your log files, which makes it easier to load the files in R.

```
bdsky_trace <- beastio::readLog(bdsky_777.log, burnin=0.1)
```

With the trace loaded as an `mcmc` object from the `coda` package we can use `coda` functions to investigate the trace and check convergence (Figure 21).

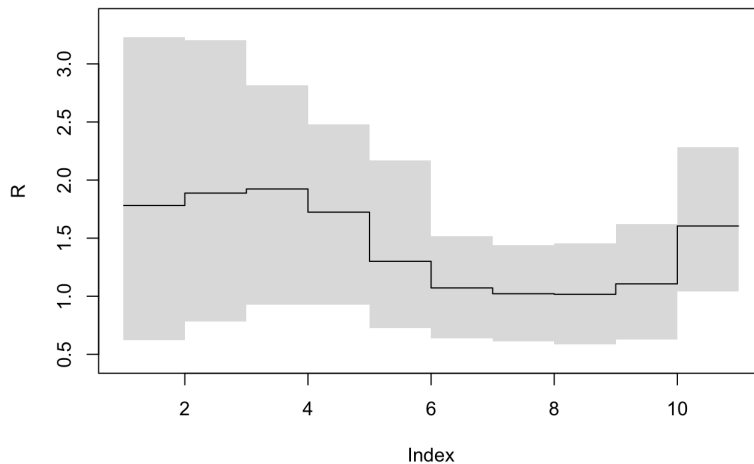


Figure 22: The HPDs of ' R_e ' (equivalent to the previous figure from Tracer).

Next we can extract the R_e parameter values for the Alpha alignment and their HPDs. We have to use a regular expression to select only the R_e parameters and not their endtimes.

```
Re_alpha <- beastio::getLogFileSubset(bdsky_trace, "ReSPEpi.alphai[0-9]+$")
Re_alpha_hpd <- t(beastio::getHPDMedian(Re_alpha))
```

We can plot the raw R_e HPD intervals. This is equivalent to the output in Tracer (Figure 22).

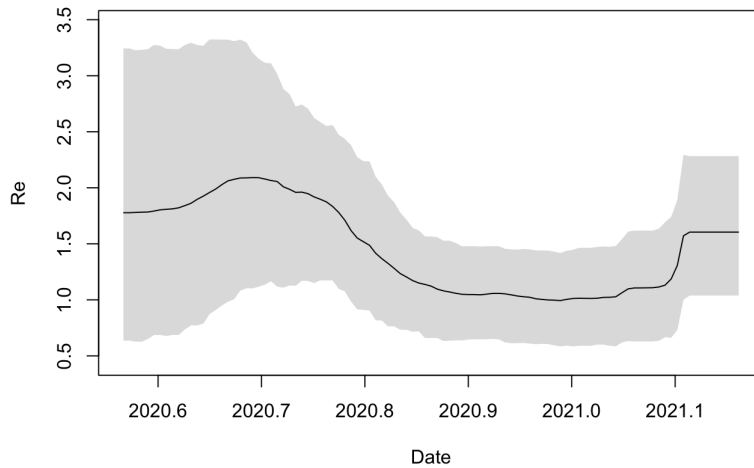
```
plotSkyline(1:10, Re_alpha_hpd, type='step', ylab="R")
```

In order to plot the smooth skyline we have to marginalise our R_e estimates on a regular timegrid and calculate the HPD at each gridpoint. It is usually a good idea to use a grid with more cells than the dimension of R_e . To do this we first calculate the marginal posterior at every time of interest using the function `gridSkyline()` and then calculate the HPD for each of the finer time intervals. The times to grid the skyline on (`gridTimes_alpha`), refers to years in the past. Since we set the change times to be distributed equally between the process origin and the present we need to use the origin as an anchor point.

```
origin_alpha <- bdsky_trace[, "originBDMMPPrime.alpha"]
gridTimes_alpha <- seq(0, median(origin_alpha), length.out=params$gridsize)

Re_alpha_gridded <- mcmc(bdskytools::gridSkyline(Re_alpha, origin_alpha, gridTimes_alpha))
Re_alpha_gridded_hpd <- t(getHPDMedian(Re_alpha_gridded))
```

Now we are ready to plot the smooth skyline (Figure 23).

Figure 23: The smooth R_e skyline.

```
times <- lubridate::decimal_date(enddate)-gridTimes_alpha
plotSkyline(times, Re_alpha_gridded_hpd, xlab="Date", ylab="Re")
```

We can plot the gridded R_e skyline (not its HPDs) for a few of the MCMC samples to see what it really looks like as the Markov chain samples parameters. Note that the intervals overlap between different posterior samples. This is because the origin estimate is different in each of the plotted samples. As we add more samples to the plot we start to see the smooth skyline appear (Figure 24).

```
plotSkyline(times, Re_alpha_gridded, type='steplines', traces=1,
  col=cols$blue,ylims=c(0,3.5), xlab="Time", ylab="R", main="1 random sample"<-
)
plotSkyline(times, Re_alpha_gridded, type='steplines', traces=10,
  col=set_alpha(cols$blue,0.5),ylims=c(0,3.5), xlab="Time", ylab="R", <-
  main="10 random samples")
plotSkyline(times, Re_alpha_gridded, type='steplines', traces=100,
  col=set_alpha(cols$blue,0.5),ylims=c(0,3.5), xlab="Time", ylab="R", <-
  main="100 random samples")
plotSkyline(times, Re_alpha_gridded, type='steplines', traces=1000,
  col=set_alpha(cols$blue,0.1),ylims=c(0,3.5), xlab="Time", ylab="R", <-
  main="1000 random samples")
```

We can do the same for the sampling proportion estimates for the Alpha alignment (Figure 25).

```
samplingProp_alpha <- beastio::getLogFileSubset(bdsky_trace, "samplingProportionSPEpi<-
.alpha[i[0-9]+$")
samplingProp_alpha_gridded <- mcmc(bdskytools::gridSkyline(samplingProp_alpha, origin<-
```

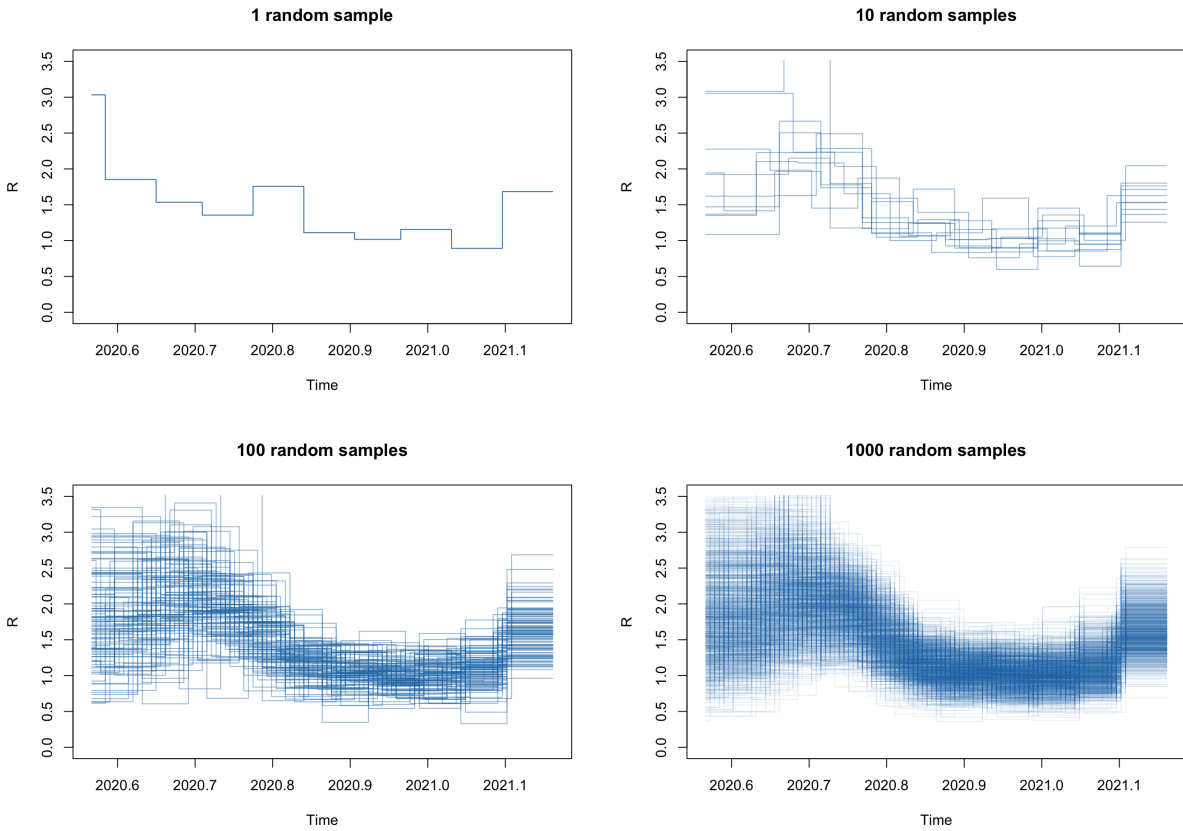


Figure 24: Increasing the number of traces plotted from 1 to 10, to 100, to 1000.

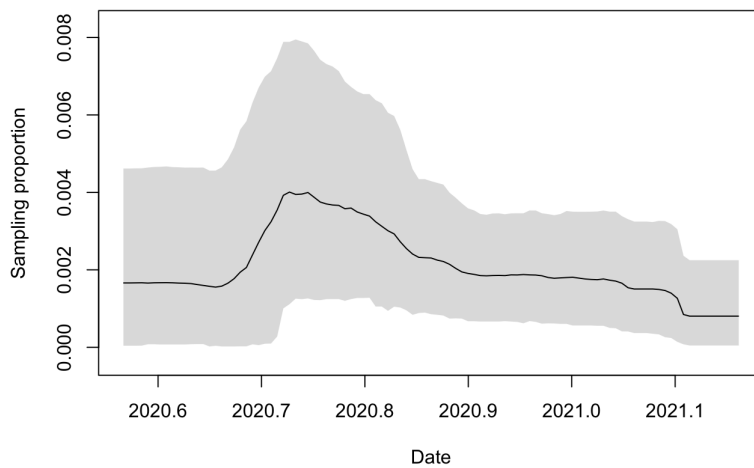


Figure 25: The smooth sampling proportion skyline.


```

_alpha, gridTimes_alpha))
samplingProp_alpha_gridded_hpd <- t(getHPDMedian(samplingProp_alpha_gridded))
plotSkyline(lubridate::decimal_date(enddate)-gridTimes_alpha, samplingProp_alpha_
  gridded_hpd,
  xlab="Date", ylab="Sampling proportion")

```

Now we can follow the same steps to also extract and plot the skylines for the background alignment. Finally, we can plot both Alpha and the background datasets on one set of axes for comparison (Figure 26).

3.8 Sampling from the prior (optional)

It is always a good idea to sample from the prior. Ideally this should be done before running the analysis, as sampling from the prior can point out issues with the model. In particular, it allows us to see how much information is encoded in the priors and model structure. For example, suppose our aim is to decide if non-pharmaceutical interventions (NPIs) were effective by looking at whether R_e dropped below one after they were implemented. If the model is constrained such that R_e is always below one after that time, the model is unable to answer that question. We can't simply look at the prior distributions we set in BEAUti to reach such insights, because priors can interact with each other in complex ways [Heled and Drummond 2011](#). The result is that the induced prior of the complete model may not be the same as the priors we set in BEAUti.

Select the **MCMC** tab.

- Set the **Chain Length** to **50'000'000**.
- Expand the **tracelog** options and set the **Log Every** parameter to **5000**.
- Expand the **treelog.t:alpha** options and set the **Log Every** parameter to **5000**.
- Expand the **treelog.t:alpha** options and set the **Log Every** parameter to **5000**.
- Check the **Sample From Prior** checkbox at the bottom of the window (Figure 27).

Now save the file as `bdsky_sampleFromPrior.xml` and run it in BEAST2.

Sampling from the prior runs the analysis without using the data. In BEAST2 this means simply masking the alignment and not calculating the tree likelihood. Since calculating the tree likelihood is usually the most time-consuming part of any analysis, sampling from the prior is usually much faster than running the analysis. However, since there is no data, it also often has difficulty mixing and needs to be run for longer.

Note that in the case of a heterochronous dataset, as we have here, sampling from the prior still uses the sampling (tip) dates in the analysis. Strictly speaking, these can also be considered part of the data. However, in our case we are also interested in seeing precisely how much information the sequences are adding on top of the collection dates.

3.9 Comparing to the prior (optional)

When comparing the posterior and prior estimates in Tracer we can see that for some parameters, such as the clock rates, the posterior is a much tighter and more peaked distribution than the prior (Figure 28). But for the Skyline parameters, the picture is much less clear. To get a better idea we have to plot the smooth skylines in R.

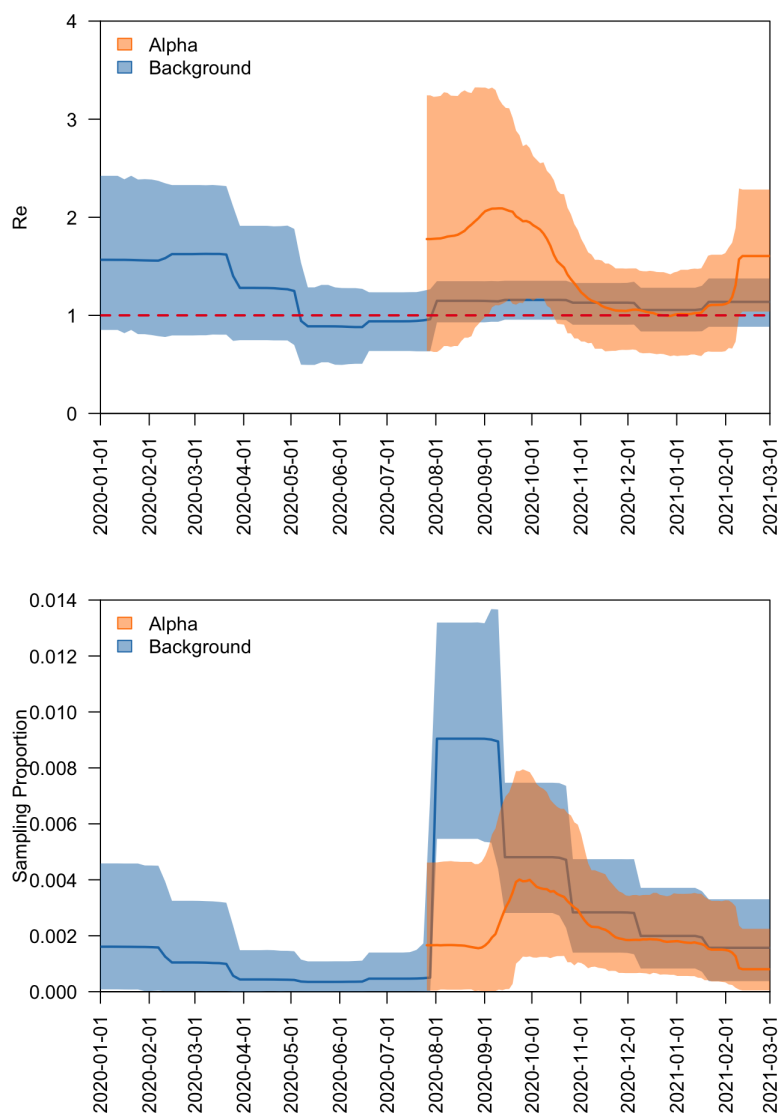


Figure 26: Plotting the skylines of the Alpha and background datasets on one set of axes.

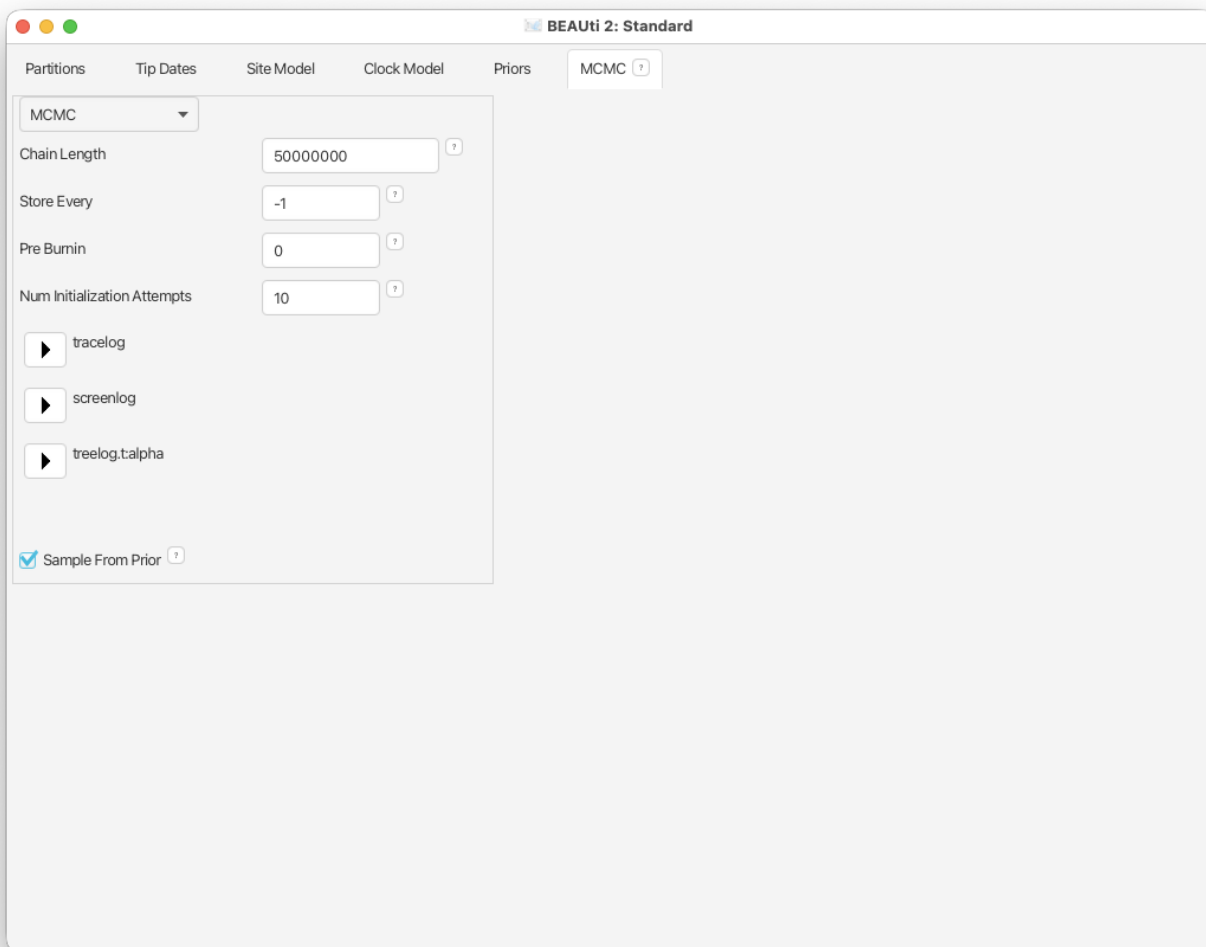


Figure 27: Sampling from the prior.

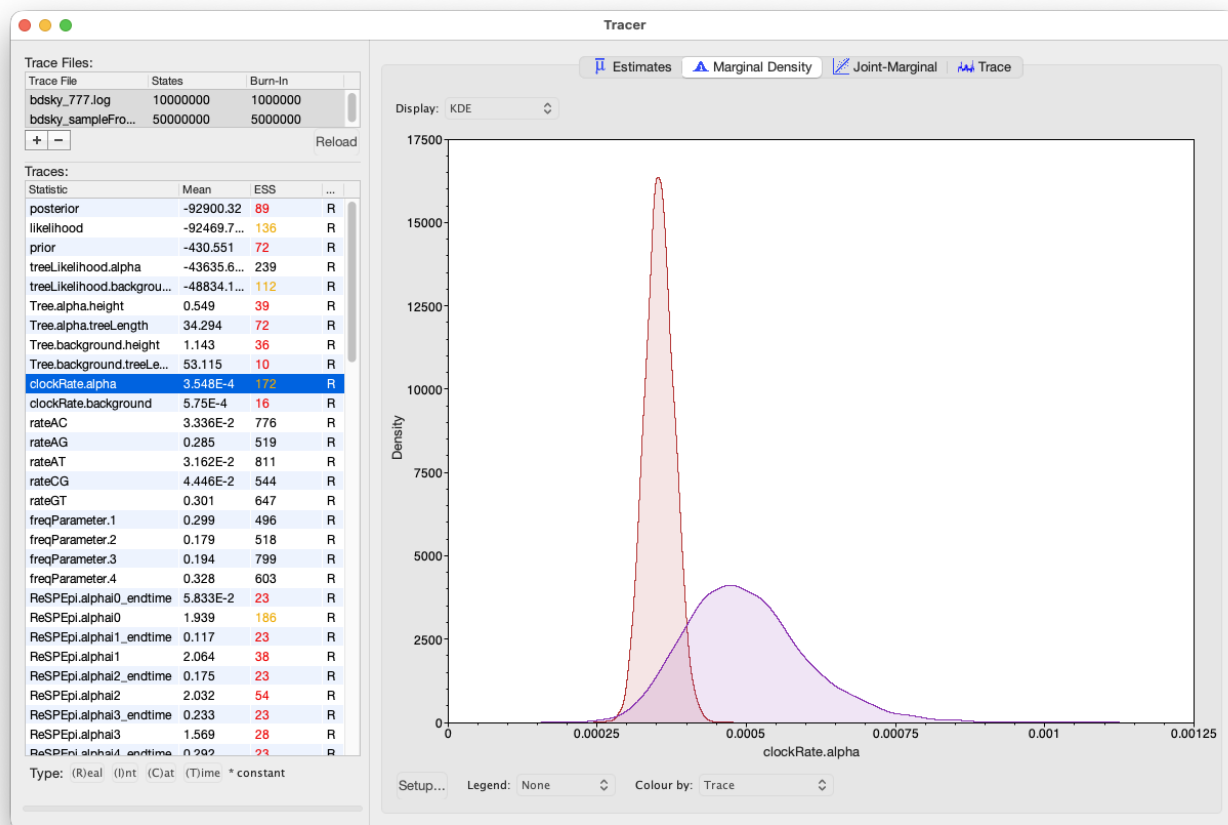


Figure 28: Comparing prior and posterior distributions in Tracer.

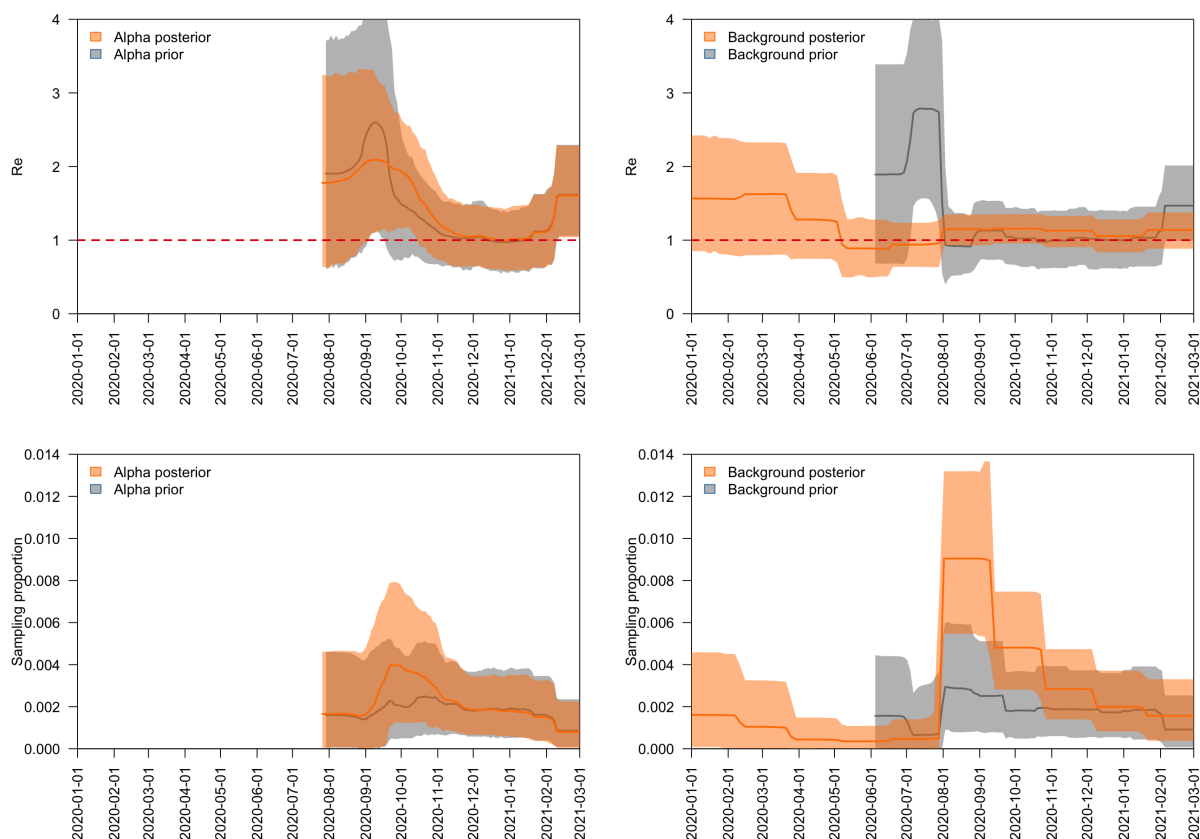


Figure 29: Comparing prior and posterior distributions of the Birth Death Skyline parameters.

We can follow the same steps as above to load the log file from `bdsky_samplingFromPrior_777.xml` into R, grid and plot the skylines, and then compare them to the posterior estimates (Figure 29).

What can you conclude from the figures above?

3.10 Limitations

Both the coalescent and the birth-death skylines assume that the population is well-mixed. That is, they assume that there is no significant population structure and that the sequences are a random sample from the population. However, if there is population structure, for instance sequences were sampled from two different villages and there is much more contact within than between villages, then the results will be biased (Heller et al. 2013). Instead a structured model should then be used to account for these biases.

For the coalescent Bayesian Skyline plot we noticed that both skylines show a constant effective population size after November 2020 (Figure 18). But this doesn't agree with what we know about the COVID-19 epidemic in the UK, where cases (especially of the Alpha VOC) increased steeply over December 2020 and then started declining steeply after the New Year with the start of a lockdown.

To understand why the Bayesian Skyline Plot can't infer these dynamics on our dataset we have to look at the inferred trees. We see that the CCD0 tree for the Alpha alignment (created in TreeAnnotator) has very long terminal branches, and many coalescent events deeper in the tree (Figure 30). This is indicative of fast exponential growth dynamics. When the population is large (in the present), the probability of two lineages coalescing is small, resulting in long terminal branches. When the population is small (in the past) the opposite is true, and branch lengths are short, resulting in many coalescent events. For the Bayesian Skyline Plot only the coalescent events are informative about N_e . Therefore, the model detects fast growth during the period of many coalescent events (October/November 2020), but has no power to detect any changes in N_e after this time. Because we only have a tiny sample from a huge epidemic wave, almost all of the coalescences in our tree occurred in October/November 2020. Had we taken a much larger sample it is possible that the model would have been able to infer more recent changes in N_e . In addition, there are variations of the coalescent model that either assumes the growth rate remains constant (i.e. N_e will keep growing) (Volz and Didelot 2018) or so-called preferential sampling coalescent models that use information from the sampling times to also inform N_e (Karcher et al. 2020; Parag et al. 2020; Karcher et al. 2016).

The Birth Death Skyline model uses both coalescent and sampling events to inform itself about the rates. However, it is apparent that our datasets are not informative enough to allow for tight estimates of R_e (Figure 26). It is also apparent that most of the information for the Alpha R_e and sampling proportion estimates come from the sampling times, and not the sequences (Figure 29). This is also true to some extent for the background alignment, although the posterior estimates deviate more from the prior.

With the background sampling proportion it is interesting to note that the estimates include 0 in their HPDs until August 1st, 2020, the time of the oldest sample in our dataset. After that, the sampling proportion jumps and then decreases again. This agrees with the situation in the UK at the time, where cases were very low during August, but started increasing from September 2020. Usually, we would want to set the sampling proportion to 0 before our oldest sample.

Because SARS-CoV-2 evolves relatively slowly (around 2 *de novo* mutations per month) sequences are not very informative over short time scales. That means larger datasets or extra information need to be included in order to make accurate inferences.

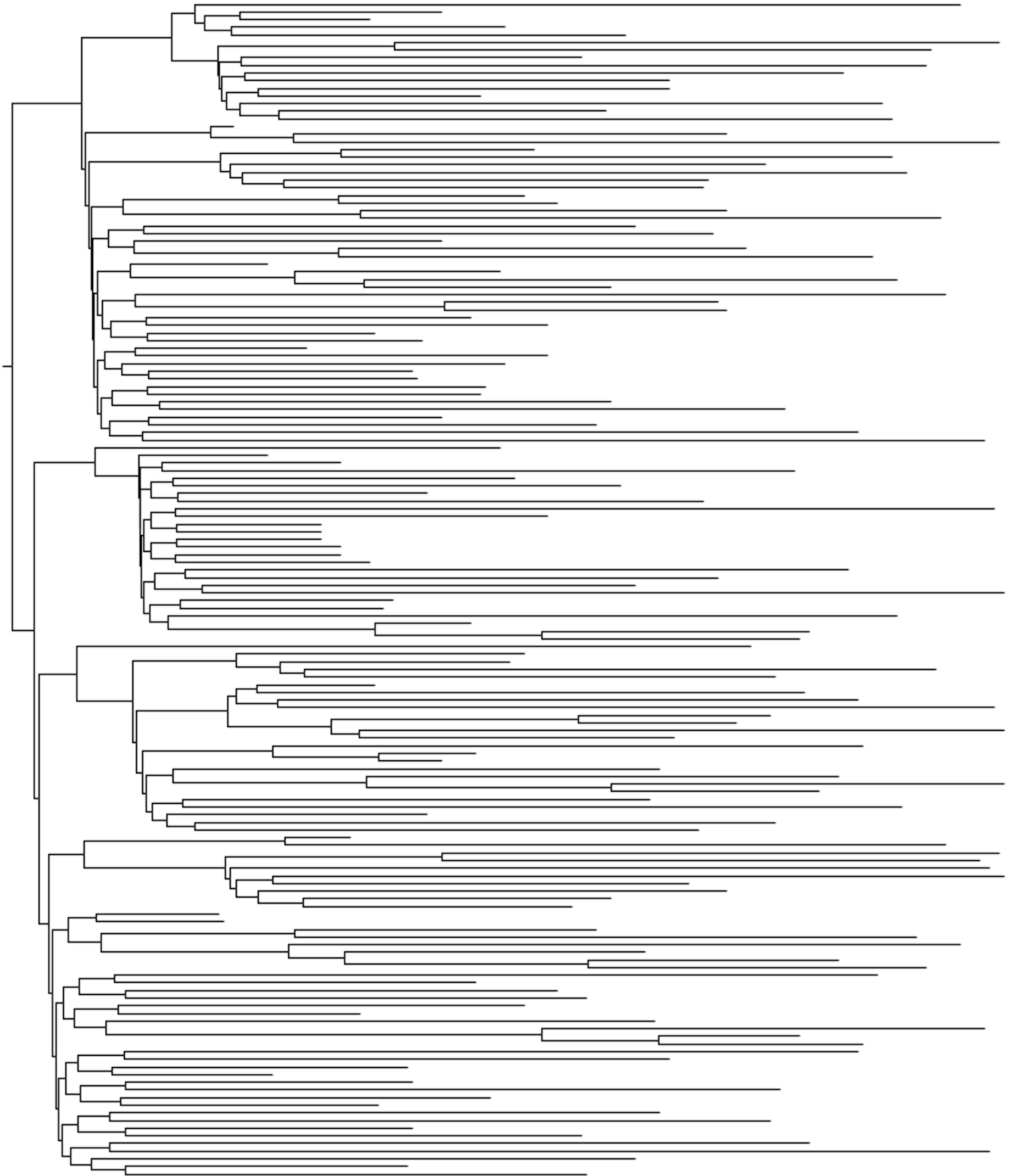


Figure 30: The CCD0 MAP tree of the Alpha alignment inferred by the Bayesian Skyline Plot.



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Version dated: June 22, 2026

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